MILITARY SPECIFICATION

RELAYS, HYBRID OR SOLID STATE, SENSORS, ESTABLISHED RELIABILITY

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for hybrid and solid state sensing relays. These relays are designed to operate over the full range from low level to power switching with contact ratings up to 10 amperes alternating current (ac) or direct current (dc). The failure rate level is established at a confidence level of 90 percent for qualification and 60 percent for maintenance of qualification based on 100,000 cycles at 125°C under rated load conditions specified herein.

1.2 Classification. Relays covered by this specification shall be of the following types and classes, as specified.

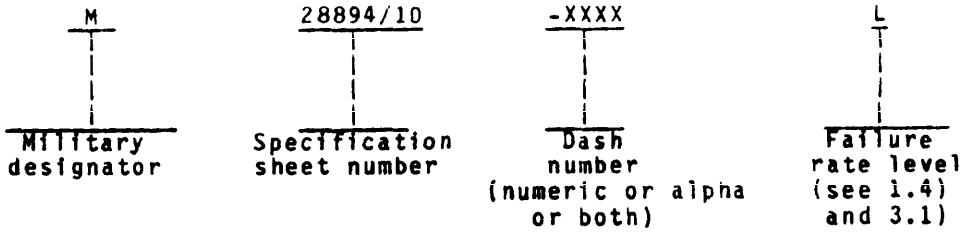
Type

I	-	AC power monitoring sensors
II	-	DC voltage sensors
III	-	AC voltage sensors
IV	-	Frequency sensors
V	-	Phase sequence sensors

Class

A	-	Hybrid (integral electromagnetic relay qualified to MIL-R-39016)
B	-	Hybrid (integral established reliability relay qualified to MIL-R-6106)
C	-	Solid state

1.3 Military part number. The military part number shall consist of the letter "M", the basic number of the of the specification sheet, an assigned dash number (see 3.1), and a suffix letter designating failure rate level (see table I) as shown in the following example:



Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Space and Naval Warfare Systems Command, ATTN: ELEX 5043, Washington, DC 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Failure rate (FR) level designation. The failure rate level designation is shown in table I (see 4.5).

TABLE I. Failure rate level designation.

Failure-rate-level designation	Failure-rate-level (percent per 10,000 cycles)
L	3.0
M	1.0
P	0.1
R	0.01

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

J-W-1177 - Wire, Magnet, Electrical.
L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
NN-P-71 - Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial).
QQ-S-571 - Solder, Tin Alloy: Tin-lead Alloy; and Lead Alloy.
QQ-S-781 - Strapping, Steel, and Seals.
ZZ-R-765 - Rubber, Silicone.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-I-10 - Insulating Materials, Electrical, Ceramic, Class L.
MIL-M-14 - Molding Plastics and Molded Plastic Parts, Thermosetting.
MIL-C-17 - Cable, Radio Frequency, Flexible and Semirigid, General Specification for.
MIL-P-116 - Preservation, Methods of.
MIL-P-997 - Plastic Material, Laminated, Thermosetting, Electrical Insulation, Sheets, Glass Cloth, Silicone Resin.
MIL-W-5088 - Wiring, Aerospace Vehicle.
MIL-R-5757 - Relay, Electromagnetic, General Specification For.
MIL-R-6106 - Relay, Electromagnetic.
MIL-P-15037 - Plastic Sheet, Laminated, Thermosetting, Glass Cloth, Melamine Resin.
MIL-P-15047 - Plastic Sheets, Laminated Thermosetting, Nylon Fabric Base, Phenolic Resin.
MIL-S-19500 - Semiconductor Devices, General Specification for.
MIL-M-38510 - Microcircuits, General Specification for.
MIL-R-39016 - Relays, Electromagnetic, Established Reliability, General Specification for.
MIL-G-45204 - Gold Plating, Electro Deposited.

(See supplement 1 for list of associated specifications.)

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STANDARDS

FEDERAL

FED-STD-H28	-	Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-143	-	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-147	-	Palletized Unit Loads.
MIL-STD-202	-	Test Methods for Electronic and Electrical Component Parts.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Emission and Susceptibility, Requirements for the Control of Electromagnetic Interference.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-690	-	Failure Rate Sampling Plans and Procedures.
MIL-STD-704	-	Aircraft Electric Power Characteristics.
MIL-STD-790	-	Reliability Assurance Program for Electronic Parts.
MIL-STD-794	-	Part and Equipment Procedures for Packaging and Packing of. Specifications.
MIL-STD-883	-	Test Methods and Procedures for Microelectronics.
MIL-STD-1285	-	Marking of Electrical and Electronic Parts.
MIL-STD-1549	-	Common Termination System for Electrical and Electronic Parts.
MS20659	-	Terminal, Lug, Crimp Style, Copper, Uninsulated, Ring Tongue, Type I, Class 1.
MS25036	-	Terminal, Lug, Crimp Style, Copper Insulated, Ring Tongue, Bell Mouthed, Type II, Class 1.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y32.2 - Graphic Symbols for Electric and Electronic Diagrams.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheets, the latter shall govern.

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3.2 Qualification. Relays furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.5 and 6.3). Unless acquired from the manufacturer or his authorized distributor listed or approved for listing on the qualified products list, parts furnished under this specification shall not be considered as having met the requirements of this specification.

3.3 Materials. The materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the relays to meet the performance requirements of this specification. Materials used shall be self-extinguishing, and shall not support combustion, give off noxious gases in harmful quantities, give off gases in quantities sufficient to cause explosion of sealed enclosures, cause contamination of the contacts or other parts of the relay, or form current-carrying tracks when subjected to any of the tests specified herein. The selection of materials shall be such as to provide maximum shelf life. Acceptance or approval of any constituent material shall not be construed as a guaranty of acceptance of the finished product.

3.3.1 Plastic. Laminated plastic material shall conform to MIL-P-997, MIL-P-15037, or MIL-P-15047. Molded plastic material shall conform to MIL-M-14. Cotton-filled or wood-flour-filled materials shall not be used.

3.3.2 Ceramic. Ceramic insulating material shall conform to MIL-I-10, grade L422 or higher. Ceramic use for external surfaces shall be glazed.

3.3.3 Fungus-resistant. Materials used in the construction of relays shall be fungus inert (see requirement 4 of MIL-STD-454).

3.3.4 Metals. Metals shall be of a corrosion-resistant type or shall be plated or treated to resist corrosion, except that zinc, cadmium, or unfused tin plating shall not be used internally. Zinc or cadmium shall not be used externally. Unfused tin plating shall have a plating thickness of 200 microinches minimum if used for external parts.

3.3.4.1 Dissimilar metals. When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metal in contact, which tends toward active electrolytic corrosion (particularly brass, copper, or steel used in contact with aluminum or aluminum alloy), is not acceptable. However, metal spraying or metal plating of dissimilar base materials to provide similar or suitable abutting surfaces is permitted. Dissimilar metals shall be as defined in 6.6 through 6.6.4, inclusive. In hermetic seals, the 0.25 volt difference between the header material and the housing material is not applicable.

3.3.5 Magnet wire. Magnet wire shall conform to J-W-1177.

3.3.6 Rubber. Rubber shall conform to ZZ-R-765.

3.3.7 Mercury. The use of mercury or mercury compounds is prohibited.

3.3.8 Magnesium. The use of magnesium or magnesium alloys is prohibited.

3.4 Design and construction. Relays shall be of the design, construction, weight, and physical dimensions specified (see 3.1).

3.4.1 Case. The case shall not be electrically connected to the contacts or coil; however, it may be used as part of the magnetic circuit.

3.4.1.1 Case Grounding. When specified (see 3.1), means for connecting the relay case to ground shall be provided.

3.4.2 Contacts. Contacts shall have load ratings and arrangements (see figure 1 and MIL-STD-1285) as specified (see 3.1) and shall be capable of carrying the maximum rated current continuously as well as making and breaking the specified current under all environmental conditions specified herein.

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3.4.3 Coils (class A and B relays only). Coils of class A and B relays shall be adquately insulated electrically from the contacts and the case. The coils shall function over the operational voltage range as specified (see 3.1) under all environmental conditions as specified herein and in the applicable detail specification sheet (see 3.1).

3.4.3.1 Terminal identification. When specified (see 3.1), a bead of contrasting color shall be used to designate the X1 (positive, if applicable) terminal (see figure 1 and MIL-STD-1285).

3.4.4 Circuit diagram. The circuit diagram as specified (see 3.1), shall be a terminal view. Circuit symbols shall be in accordance with ANSI Y32.2. For relays without an orientation tab, the circuit diagram as specified (see 3.1), shall be oriented so that when the relay is held with the circuit diagram right side up as shown (see 3.1), and rotated away from the viewer about a horizontal axis through the diagram until the header terminals face the viewer, then each terminal shall be in the location shown in the circuit diagram.

3.4.5 Terminals (electrical). Terminals shall be in accordance with the following and as specified (see 3.1). No rotation or other loosening of a terminal, or any fixed portion of a terminal, shall be caused by material flow or shrinkage, or any mechanical forces specified in table II involved in connection or disconnection, throughout the life of the relay. Unless otherwise specified (see 3.1), terminals shall be marked or be adequately described by the circuit diagram in accordance with figure 1 and MIL-STD-1285.

TABLE II. Strength of terminals (static value of pull and torque).

Thread size	Force in pounds	Torque in pound-inches
.112-40	5	4.4
.138-32	30	10.0
.164-32	35	20.0
.190-32	40	32.0
.190-24	40	35.0
.25-28	50	75.0
.3125-24	70	100.0
.375-24	100	150.0
.4375-20	100	150.0
.50-20	100	150.0

3.4.5.1 Stud terminals (threaded). Unless otherwise specified (see 3.1), stud terminals shall be supplied with two flat washers, one nut, and one split ring type lockwasher. A minimum of three complete threads shall remain above the nut when it is backed off three complete turns from a position with all parts tightened in place.

3.4.5.1.1 Stud terminal seat. For threaded terminals, each terminal shall have a terminal seat that shall provide the normal current-conducting path. The diameter of the seat shall be equal to, or greater than, the diameter across the corresponding MS20659 or MS25036 lug designed for the particular current and stud or screw size, or never less than the area necessary to assure that the current density shall not exceed 1,000 amperes per square inch. The seat area does not include the cross sectional area of the stud.

3.4.5.1.2 Strength of stud terminals. Stud terminals shall be designed to withstand the static value or pull and torque specified in table II (see 3.35).

3.4.5.2 Solder-lug, socket plug-in, printed-wiring, and wire lead terminals.

3.4.5.2.1 Solder-lug terminals. Unless otherwise specified (see 3.1), solder-lug terminals may be of any shape, and shall be capable of being readily soldered. Solder-lug terminals shall be designed to accommodate two 20 stranded wires with 19 strands for relays rated for 2 amperes and less, and two wires, each rated to carry the maximum rated current of the relay, for relays rated above 2 amperes (see MIL-W-5088 for current carrying capability of wire). Input terminals to the sensing

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circuitry shall be designed to accommodate two 20 stranded wires with 19 strands. Unless otherwise specified (see 3.1), solder-lug terminals shall not be gold plated.

3.4.5.2.2 Socket plug-in terminations. Socket plug-in terminations shall conform to the arrangements or dimensions necessary for proper mating with the associated connectors or sockets as specified (see 3.1). The mounting arrangement of the relay and its corresponding socket shall be so designed that the entire weight of the relay will be suspended and the stability of its mounting will be provided by an auxiliary mounting means other than the electrical terminals of the socket. Relays with plug-in terminals shall have electrical and environmental tests of section 4 performed with the appropriate or specified socket or connector assembled to the relay.

3.4.5.2.3 Wire leads. Wire leads shall be as specified.

3.4.5.2.4 Strength of solder-lug, socket, plug-in, printed wiring, and wire lead terminals. The relay terminals shall be designed to withstand the applicable terminal strength performance procedure specified in 3.35.

3.4.5.3 Terminal finish. Finish of terminals shall provide a good electrical contact and meet the performance requirements specified herein. All terminals used for external soldered connections shall be tin plated or coated with composition Sn40 to Sn70 solder conforming to QQ-S-571 to facilitate soldering. Socket plug-in terminals shall be gold plated in accordance with MIL-G-45204, type II, class I.

3.4.6 Mounting means (see 3.1).

3.4.6.1 Socket. Socket plug-in relays shall be so designed that the weight of the relay will be supported and the stability of the mounting will be provided by means other than the terminals.

3.4.6.2 Bracket. Mounting brackets shall be an integral part of the relay case or shall be securely attached thereto in a manner to prevent any movement between the relay and the mounting bracket.

3.4.6.3 Mounting studs. Mounting studs shall be as specified (see 3.1). No rotation, loosening, or deformation of fixed portions shall occur because of material flow or any mechanical forces involved in installation or removal of the relay. The mounting studs shall withstand for 1 minute without damage, the static values of pull and torque specified in table III (see 3.35). Unless otherwise specified (see 3.1), each stud shall be supplied with two flat washers, one nut, and one lockwasher. For direct Government orders, the hardware shall be assembled on the stud.

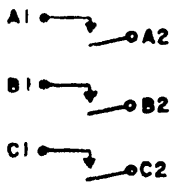
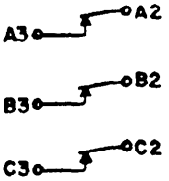
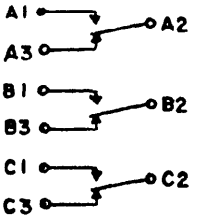
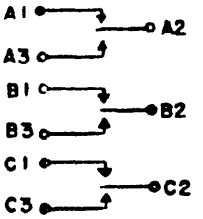
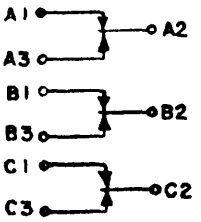
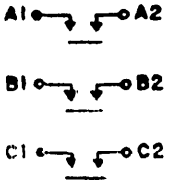
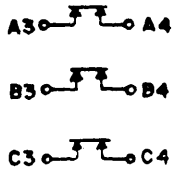
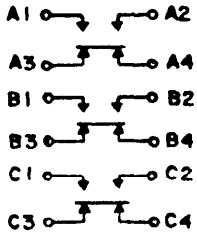
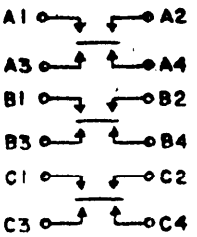
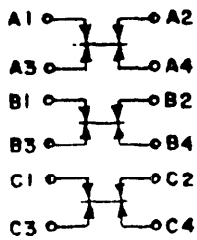
TABLE III. Strength of mounting studs (static values of pull and torque.

Thread size	Force in pounds	Torque in pound-inches
.112-40	5	10
.138-32	25	18
.164-32	35	37
.190-32	50	60
.25-28	60	100
.3125-24	80	160
.375-24	115	275
.4375-20	140	475

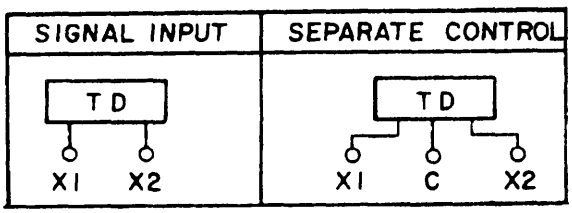
3.4.7 Dimensions and weight. Relay physical dimensions and weight shall be as specified (see 3.1).

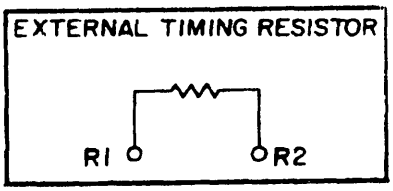
3.4.8 Solder. Solder shall have a minimum softening point 55°C above the maximum temperature rating for the relay. Wherever suitable for the purpose, compositions Sn70, Sn60, and Sb5, conforming to QQ-S-571, shall be used. Solder of other compositions may be used, provided they possess superior characteristics and meet the requirements of this specification.

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CONTACT ARRANGEMENTS, SYMBOLS AND TERMINAL MARKING (SEE NOTE)					
	SINGLE THROW		DOUBLE THROW		
SINGLE BREAK	FORM "A" NORMALLY OPEN	FORM "B" NORMALLY CLOSED	FORM "C" (TWO POSITION)	FORM "K" SPDT 3 POS CENTER OFF	FORM "M" 3 POS CENTER ON
					
	FORM "X"	FORM "Y"	FORM "Z"	FORM "KK"	FORM "MM"
DOUBLE BREAK					

NOTE: CONTACTS ARE SHOWN WITH COIL(S) DE-ENERGIZED.





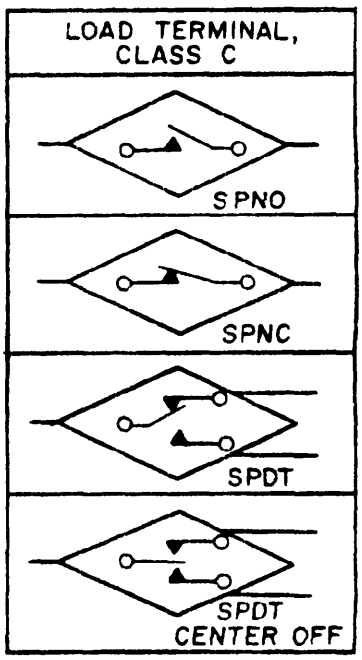


FIGURE 1. Terminal markings.

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3.4.8.1 Soldering processes. Soldering shall be so executed that both a positive electrical and strong mechanical connection is assured. Electrical joints shall be mechanically secured before soldering and shall not depend upon solder for mechanical strength.

3.4.9 Microelectronic components. Microelectronic components shall be selected in accordance with the applicable requirements of MIL-STD-454. Components selected shall be such that the relays shall meet the performance requirements and product characteristics specified herein. After qualification, any change of parts or material shall be submitted to the Government qualifying activity for approval.

3.4.9.1 Discrete semiconductor devices. Discrete semiconductor devices including diodes, zener diodes and transistor devices shall be selected from MIL-S-19500 whenever possible. Non-standard discrete semiconductor devices may be used in level L thru R relays, provided that these devices are tested to meet the screening requirements of MIL-S-19500 for level JANTX.

3.4.10 Selection of military specifications and standards. Specifications and standards for all parts, materials, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, as applicable, except as provided in the following paragraph.

3.4.10.1 Standards parts. Standard mounting and terminal hardware (i.e., screws, lockwashers, etc.) (MS and AN) shall be used wherever practicable. Nonstandard equivalents may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration.

3.4.11 Integral electromagnetic relay (class A and B relays). The integral electromagnetic relay (mounting means excepted) used in hybrid relays covered by this specification shall have been tested and qualified, with the exception of the headers and coils which shall meet the requirements of 3.4 and 3.4.3 respectively to the following specifications, as applicable (see 3.1):

Class	
A	MIL-R-39016
B	MIL-R-6106

3.5 Screening. When relays are tested as specified in 4.9.2 (for class A and B relays), during cycling, the contact miss detectors monitoring level shall be less than 100 ohms. Unless otherwise specified (see 3.1), all relay classes shall have a final insulation resistance measurement of 100 megohms or greater.

3.6 Solderability. When relays are tested as specified in 4.9.3, the dipped surface of solid wire-lead and pin terminals shall be at least 95 percent covered with a continuous new solder coating. The remaining 5 percent may contain only small pinholes or rough spots; these shall not be concentrated in one area. Bare base metal where the solder dip failed to cover the original coating is an indication of poor solderability, and shall be cause for failure. For solder-lug terminals greater than 0.045 inch (1.14 mm) in diameter, 95 percent of the total length of fillet, which is between the standard wrap wire and the terminal, shall be tangent to the surface of the terminal being tested, and shall be free of pinholes, voids, etc. A ragged or interrupted line at the point of tangency between the fillet and the terminal under test shall be considered a failure.

3.7 Seal (see 4.9.4).

3.7.1 Hermetically-sealed (potted) relays (i.e., relays that employ metal-to-metal, metal-to-glass, or metal-to-ceramic sealing but are filled with potting). When relays are tested as specified in 4.9.4.1, there shall be no evidence of leakage.

3.7.1.1 Nonhermetically-sealed relays. When relays are tested as specified in 4.9.4.2, there shall be no evidence of leakage.

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3.8 Insulation resistance. When relays are tested as specified in 4.9.5, the insulation resistance shall be 100 megohms or more, unless otherwise specified (see 3.1). After the high level life tests, the insulation resistance shall be 50 megohms or more.

3.9 Dielectric withstanding voltage. When tested as specified in 4.9.6, relays shall withstand the test voltage specified without damage, and there shall be no leakage current in excess of 100 microamperes (μA) nor evidence of damage due to arcing (air discharge), flashover (surface discharge), or insulation breakdown (puncture discharge). After high level life tests, the dielectric withstanding voltage shall be at least 75 percent of the initial sea level value (see 3.1).

3.10 Electrical characteristics (see 4.9.7).

3.10.1 Static contact resistance (or voltage drop). When relays are tested as specified in 4.9.7.1, unless otherwise specified (see 3.1) the static contact resistance shall not exceed the values specified in table IV.

TABLE IV. Static contact resistance (or voltage drop).

Contact rating or relay type	Load conditions	Maximum allowable static contact resistance (or voltage drop)	
		Before life	During and after life
High level (more than 2 amperes but not in excess of 10 amperes)	Rated current at 6 or 28 V dc or peak ac	100 mV	200 mV
High level (2 amperes or less)	Rated current at 6 or 28 V dc or peak ac	0.05 ohm	0.10 ohm
Intermediate current	100 ±5 mA at 28 V dc or peak ac	0.05 ohm	3.0 ohms
Low level	10 mA maximum at 6 V open-circuit (maximum dc or peak ac)	0.05 ohm	0.15 ohm

3.10.2 Pickup time and dropout time (class A and B relays only) (see 4.9.7.2). When relays are tested as specified in 4.9.7.4 the pickup time and dropout time shall be 50 milliseconds maximum, unless otherwise specified (see 3.1). In multipole relays all poles of each relay shall function simultaneously within 1 millisecond (ms). Double throw contacts shall show no evidence of a normally open contact being closed concurrently with a normally closed contact.

3.10.3 Contact dynamic characteristics (class A and B relays only) (see 4.9.7.3).

3.10.3.1 Contact bounce (applicable to failure rate level L). When relays are tested as specified in 4.9.7.3.1, the duration of the contact bounce shall not exceed 1.5 ms unless otherwise specified (see 3.1). After life test, the duration of contact bounce shall be as specified (see 3.1).

3.10.3.2 Contact stabilization time (applicable to failure rate levels M, P and R). When relays are tested as specified in 4.9.7.3.2, the time to reach and maintain a static contact resistance state shall be as specified (see 3.1).

3.11 Input current. When relays are tested as specified in 4.9.8, the input and control current, if applicable, exclusive of the output load shall not exceed the value specified (see 3.1).

3.12 Leakage current (class C relays). When relays are tested as specified in 4.9.9, the leakage current shall not exceed the maximum value specified (see 3.1).

3.13 Waveform distortion (class C, ac relays). When tested as specified in 4.9.10, the output waveform distortion shall not exceed the limits specified (see 3.1).

3.14 DC offset voltage (class C, ac relays). When relays are tested as specified in 4.9.11, the dc offset voltage shall not exceed the value specified (see 3.1).

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3.15 Current range (class C, ac relays). When tested as specified in 4.9.12, unless otherwise specified, the dc offset voltage and waveform distortion shall not exceed specified values for all load current between 1 percent and 100 percent of relay rating.

3.16 Zero crossover (class C, ac relays). When tested as specified in 4.9.13, relay turn-on shall occur at the zero voltage crossover ±6 volts, and relay turn-off shall occur at the zero current crossover ± rated load x 10^{-2} (maximum). The relay shall turn-off at the opposite half-cycle from turn-on.

3.17 Bias current (when applicable) (class C relays). When relays are tested as specified in 4.9.14, the bias current shall not exceed the value specified (see 3.1).

3.18 Maximum deactuation voltage (class C relays). When relays are tested as specified in 4.9.15, the maximum deactuation voltage shall not be greater than that specified (see 3.1).

3.19 Transients (see 4.9.16).

3.19.1 Voltage (surge). When tested as specified in 4.9.16.1, the relay shall withstand the application of transient voltage without damage or effect on the delay period (beyond specified tolerances) within the timing cycle specified (see 3.1).

3.19.2 Susceptibility (spike). When tested as specified in 4.9.16.2, the relay shall not malfunction or be damaged. Following the test, the timing cycle and recycle time shall be as specified in 3.20 and 3.21, respectively.

3.19.3 Susceptibility, extraneous voltage. When tested as specified in 4.9.16.3, the relay shall not operate over the specified temperature range (see 3.1) when 25 percent of the nominal control voltage (see 3.1) is applied to the control terminals.

3.19.4 Self-generated (spike). When the relay is tested as specified in 4.9.16.4, it shall not generate transient voltages in excess of those specified (see 3.1).

3.20 Timing cycle. When relays are tested as specified in 4.9.17, all switching circuits shall make positive contact or open as applicable. The timing cycle, period, interval, or sequence shall be within the tolerance specified (see 3.1). There shall be no momentary change in output state if the input voltage is removed prior to the end of the timing cycle.

3.21 Recycle time. When tested as specified in 4.9.18, the relay shall repeat normal operation and the timing cycle shall be as specified in 3.20.

3.22 Reverse polarity (dc operated relays). When tested as specified in 4.9.19, the relay shall not operate, start timing, or be damaged.

3.23 Electromagnetic interference. When tested as specified in 4.9.20, the relay shall not create nor be susceptible to electromagnetic interference in excess of the limits specified in MIL-STD-461, class ID.

3.24 Crosstalk (when specified, see 3.1). When relays are tested as specified in 4.9.21, the attenuation shall be a minimum of 20 dB, unless otherwise specified.

3.25 Capacitance (when specified, see 3.1). When relays are tested as specified in 4.9.22, the capacitance shall not exceed the values specified (see 3.1).

3.26 Continuous current. When tested as specified in 4.9.23, relays shall remain operative and there shall be no damage. Following the test, the insulation resistance, dielectric withstanding voltage, static contact resistance (or voltage drop), timing cycle, and recycle time shall be as specified in 3.8, 3.9, 3.10.1, 3.20 and 3.21, respectively.

3.27 Resistance to soldering heat (not applicable to plug-in relays). When tested as specified in 4.9.24 there shall be no damage to the relay, and the insulation resistance, dielectric withstanding voltage, timing cycle at 23°C, and recycle time shall be as specified in 3.8, 3.9, 3.20 and 3.21, respectively.

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3.28 Salt spray (corrosion). When subjected to the salt spray test of 4.9.25, and following washing in cold running tap water and drying for 6 hours at 65°C, the relay shall show no evidence of corrosion sufficient to impair the operation of the relay. Following this test, the timing cycle and recycle time shall be as specified in 3.20 and 3.21.

3.29 Thermal shock. When tested as specified in 4.9.26, there shall be no damage to the relay, loosening of terminals, or cracking or flaking of glass insulation (other than crazing or chipping of the glass meniscus). Following the test, the insulation resistance, dielectric withstanding voltage, static contact resistance (or voltage drop), timing cycle, and recycle time shall be as specified in 3.8, 3.9, 3.10.1, 3.20, and 3.21, respectively.

3.30 Low temperature operation. When tested as specified in 4.9.27, there shall be no damage to the relay, loosening of terminals, cracking or flaking of glass insulation (other than crazing or chipping of the glass meniscus) or of the hermetic seal. Following the test and at the specified low temperature, timing cycle, recycle time and static contact resistance shall meet the requirements of 3.20, 3.21, and 3.10.1, respectively, and shall continue to meet timing cycle and recycle time requirements until the relay returns to room temperature.

3.31 Shock (specified pulse). When relays are tested as specified in 4.9.28 unless otherwise specified, there shall be no opening of closed contacts in excess of 10 microseconds (µs) and there shall be no closing of open contacts in excess of 1 µs and no evidence of mechanical or electrical damage.

3.32 Vibration. When relays are tested as specified in 4.9.29, unless otherwise specified (see 3.1), there shall be no opening of closed contacts in excess of 10 µs and there shall be no closing of open contacts in excess of 1 µs. There shall be no evidence of loosening of parts.

3.33 Acceleration (unless otherwise specified, see 3.10). When relays are tested as specified in 4.9.30 the contacts shall remain in the deenergized condition with no input voltage.

3.34 Resistance to solvents. When relays are tested as specified in 4.9.31 the marking shall remain legible.

3.35 Strength of terminals and mounting studs (see 4.9.32).

3.35.1 Strength of threaded terminals and mounting studs. Relays having threaded stud type terminals or stud type mounting shall be tested to determine compliance with terminal strength design requirements specified in 3.4.6.1.2 and 3.4.7.3. When relay terminals are tested as specified in 4.9.32.1, the terminals shall not loosen or rotate, nor shall there be any other damage such as cracking or flaking of glass insulator other than crazing or chipping of the glass meniscus. There shall be no deterioration of relay performance beyond limits specified (see 3.1).

3.35.2 Strength of solder-lug, plug-in, and wire-lead terminals. When relay terminals are tested as specified in 4.9.32.2, there shall be no loosening or breakage of terminals, cracking or flaking of glass insulators other than crazing or chipping of the glass meniscus, or any other damage that would affect relay performance beyond specified limits.

3.36 Overload (see 4.9.32).

3.36.1 Overload (applicable to class A and class B) (applicable to class C when specified). When relays are tested as specified in 4.9.33.1, the voltage drop across closed contacts shall be not more than 5 percent of applied circuit load voltage and the voltage across open contacts shall be not less than 95 percent of applied circuit voltage. The case-to-ground fuse shall remain electrically continuous.

3.36.2 Class C relays (when specified). When relays are tested as specified in 4.9.33.1, there shall be no electrical failure. Following the test, the timing cycle and recycle time shall be as specified in 3.20 and 3.21, respectively.

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3.37 Life. When relays are tested as specified in 4.9.34 during cycling, the contact miss detector's monitoring level shall be less than 100 ohms for low level testing and less than 5 percent of the open-circuit voltage for high level testing and the voltage across open contacts shall be 95 percent or more of the open circuit voltage. The case to ground fuse shall remain electrically continuous. The static contact resistance during and following cycling shall not be greater than twice the initial specified contact resistance requirement. Class C relays shall be tested at each contact load specified.

3.38 Intermediate current. When relays are tested as specified in 4.9.35 during cycling, unless otherwise specified, the resistance of a closed contact shall be not more than 3 ohms and the voltage across an open contact shall be 95 percent or more of applied load voltage. After cycling, the static contact resistance shall not exceed the limits for high level relays specified after life test cycling at high temperature and at room ambient temperature.

3.39 Marking.

3.39.1 "JAN" brand. Relays acquired under this specification shall bear the "JAN" brand except that in the case of small relays, the letter "J" shall be used. The United States Government has adopted, and is exercising legitimate control over the certification marks "JAN" and "J", respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of militry specifications. Accordingly, items required to, and meeting all of the criteria specified herein and in applicable specification sheet (see 3.1) shall bear the certification mark "JAN". The "JAN" shall be placed on the first line above or below the part number. The "JAN" or "J" shall not be marked in front of the military part number. Items furnished under contracts or orders which either permit or require deviation from the conditions or requirements specified herein or in the applicable specification sheets (see 3.1) shall not bear "JAN" or "J". In the event an item fails to meet the requirements of this specification and the applicable specification sheets (see 3.1), the manufacturer shall remove the "JAN" or the "J" from the sample tested and also from all items represented by the sample. The "JAN" or "J" certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration No. 504,860 for the certification mark "JAN".

3.39.2 Identification marking (full). Relays shall be marked in accordance with method I of MIL-STD-1285 and shall include the following information:

a. Military part number (see 3.1).

b. "JAN" brand.

c. Date code.

d. Source code.

e. Lot symbol.

f. Input voltage, current, and frequency as specified (see 3.1).

g. Contact rating (the highest dc resistive load rating shall be marked) (see 3.1).

h. Circuit diagram (see 3.4.4).

i. Terminal marking (when applicable, see 3.1, 3.4.3.1 and 3.4.5).

3.39.3 Minimum marking. When space does not permit the marking specified in 3.39.2, the marking shall include, as a minimum the military part number; the "J" with the date code (example J8030); circuit diagram; and manufacturer's name or source code.

3.39.4 Failure rate level substitution. Relays qualified to lower failure (better) rates, may be substituted for higher failure rate parts. For example, a relay qualified to failure rate level "P" (0.1 percent/10,000 cycles), may be substituted for a failure rate "M" (1.0 percent/10,000 cycles), relay. Relays shall not be remarked unless specified in the contract or purchase order.

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3.40 Workmanship. Relays shall be fabricated in such a manner as to be uniform in quality, and shall be free from cracked or displaced parts, sharp edges, burrs, and other defects that will affect life, serviceability, and appearance.

3.41 Interchangeability. All parts having the same military part number shall be directly and completely interchangeable with each other with respect to installation and performance to the extent specified in the military specification sheet (see 3.1). Changes in manufacturer's number shall be governed by the drawing number requirements of DOD-STD-100.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted ot the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective materials, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Reliability assurance program. A reliability assurance program shall be established and maintained in accordance wth MIL-STD-790. Evidence of such compliance shall be verified by the qualifying activity of this specification as a prerequisite for qualification and continued qualification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

a. Qualification inspection (see 4.5).

b. Verification of qualification (see 4.6).

c. Quality conformance inspection (see 4.8).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data that the materials listed in table V used in fabricating the relays, are in accordance with the applicable referenced specifications prior to such fabrication.

TABLE V. Materials inspection.

Material	Requirement paragraph	Applicable inspection
Plastic - - - - - - - - - - - - - -	3.3.1	MIL-P-997, MIL-P-15037, MIL-P-15047, MIL-M-14
Ceramic - - - - - - - - - - - - - -	3.3.2	MIL-I-10
Integral electromagnetic relay- -	3.4.11	MIL-R-39016 for class A; MIL-R-6106 for class B
Microelectronics components - - -	3.4.9	MIL-STD-454, MIL-S-19500

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4.4 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specifed in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.4.1 Power supply. Unless otherwise specified herein, the power supply shall have no more than 10 percent regulation at 110 percent the specified test load current. A dc power supply shall have no more than 5 percent ripple voltage. An ac power supply shall be within 1 percent of the specified frequency and shall be sinusoidal with a form factor between 0.95 and 1.25.

4.4.2 Grounding. Unless otherwise specified (see 3.1), the negative side of the dc power supply shall be grounded. One side of single-phase ac power supply shall be grounded; or the neutral of 3-phase ac power supply shall be grounded, as applicable.

4.4.3 Load conditions during tests. The coils of the relay being tested shall be connected to the grounded side of their power supply; the loads of the relay being tested shall be connected to the grounded side of their power supply; and each stationary contact shall be connected to an individual load. All tests during which the contacts are loaded and being cycled, except dielectric withstanding voltage, shall be conducted with the case of the relay connected to the power supply ground or neutral through a normal blow fuse rated at 5 percent of the contact load, maximum, but not less than 0.100 ampere. For relays with nongrounded case ratings, tests for isolated-case ratings may be made with the case electrically isolated from the power supply ground.

4.4.4 Testing devices. Devices used in the testing of relays shall not load the contacts above 10 mA resistive at 6 V dc or peak ac maximum open circuit unless otherwise specified herein.

4.4.5 Mounting relays for ambient temperature tests. When the relays are subjected to the tests specified in 4.9.36 (intermediate current), they may be mounted on a heat sink in accordance with the following:

a. Each relay may be attached by its normal mounting means to a 1/16 inch thick minimum, flat aluminum plate heat sink. The heat sink shall be designed to place every relay in the center of its own square space whose total surface area (both sides) is eight times the outside surface area of the relay, excluding mounting. Relays without mounts shall be held to the heat sink with a metal strap 1/4 inch wide by 0.015 inch (6.38 mm) maximum thickness. The heat sink assembly shall be suspended by twine or other nonheat conducting material in a plane parallel to the normal air flow in the oven. The leads shall not constitute a heat sink.

b. Chamber temperature shall be controlled to maintain the temperature at the specified ambient extremes (see 3.1).

4.4.6 Methods of inspection. Application of coil power to relays under test shall be such that plus polarity is applied to the color coded terminal when applicable; or to the lower numbered terminal when color coding is not used.

4.4.7 Reliability requirements. The reliability requirements specified herein are in accordance with MIL-STD-690. The confidence level for qualification is 90 percent and the confidence level for maintenance of qualification is 60 percent.

4.5 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.5.1 Qualification. Qualification shall be granted at the "L" or "M" failure rate initially and shall be based on results of the qualification inspection specified in table VI.

4.5.1.1 Sample size. The number of relays to be subjected to qualification inspection shall be as specified in table VI. The sample shall be selected from a production run and shall be produced with equipment or procedures normally used in production. The qualification sample shall be as defined in table VI.

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4.5.2 Inspection routine. Sample units shall be subjected to the qualification inspection outlined in table VI, in the order shown, except that groups II through VI may be conducted concurrently. All sample units shall be subjected to the inspections of group I. These sample units shall then be divided into four groups as specified in table VI and subjected to the inspection specified for their particular group.

4.5.3 Failures. Failures in excess of those allowed in table VI shall be cause for refusal to grant qualification approval.

4.5.4 FR qualification. FR qualification shall be in accordance with the general and detailed requirements of MIL-STD-690 and the following details:

a. Procedure I: Qualification at the initial FR level.
Level L or M as applicable (corresponding FR percentage) of FRASP: 90 shall apply. Sample units shall be subjected to the qualification inspection specified in group IV, table VI.

b. Procedure II: Extension of qualification to lower FR levels.
To extend qualification to a lower FR level (see table VII), data from two or more styles of similar construction may be combined.

c. Procedure III: Maintenance of FR level qualification.
Maintenance period of table VIII shall apply. Regardless of the number of production lots produced during this period, the specified number of life test unit operations shall be accumulated to maintain qualification. The manufacturer shall preselect the sampling plan to be used during the maintenance period from table VIII. In the event failures occur exceeding the sampling plan selected (C = number), the failure rate of the product will be increased one level or the product removed from the qualified products list. The qualifying activity shall be promptly informed of the failure(s).

4.6 Verificaton of qualification. Every 6 months the manufacturer shall compile a summary of results of quality conformance inspections and (where applicable) extended FR test data, in the form of a verification of qualification report, and forward it to the qualifying acitivity as the basis of continued qualification approval. In addition to the periodic submission of FR test data, the manufacturer shall immediately notify the qualifying activity whenever the FR data indicates that the manufacturer has failed to maintain his qualified FR level. Continuation shall be based on evidence that, over the 6-month period, the following has been met.

a. Verification by the qualifying activity that the manufacturer meets the requirement of MIL-STD-790.

b. The manufacturer has not modified the design of the item.

c. The specification requirements for the item have not been amended so as to affect the character of the item.

d. Lot rejection for group A inspection does not exceed 10 percent or one lot, whichever is greater.

e. Requirements for groups B and C are met (where applicable).

f. The records of all FR tests combined substantiate that the "L" (3.0 percent), "M" (1.0 percent), or "P" (0.1 percent) FR levels have been maintained or that the manufacturer continues to meet the "R" (0.01 percent) FR level for which qualified, although the total component cycles of testing do not, as yet, meet the requirements of 4.5.4 (see table VIII).

If group C test requirements were not met and the manufacturer has taken corrective action satisfactory to the Government, the forwarding of the verification of qualification report may be delayed until within 30 days after completion of retesting of the group C inspections. In this case, the qualifying activity shall be notified of this condition within the time the original verification of qualification report was due. All reports shall be certified by a responsible company official and the Government inspector.

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specified in 4.9.4.2 ... when relays are tested as s

TABLE VI. Qualification inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Group I				
All relay classes				
Screening	3.5	4.9.2	All sample units	0
Visual and mechanical inspection (internal) 1/	3.1, 3.3, 3.4, and 3.40	4.9.1		
Solderability (3 sample units)	3.6	4.9.3		
Insulation resistance	3.8	4.9.5		
Dielectric withstanding voltage	3.9	4.9.6		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.10.1	4.9.7.1		
Pickup time and dropout time	3.10.2	4.9.7.2		
Contact dynamic characteristics	3.10.3	4.9.7.3		
Input current (when applicable)	3.11	4.9.8		
The following tests are applicable to class C relays only				
Leakage current	3.12	4.9.9		
Waveform distortion (ac relays)	3.13	4.9.10		
DC offset voltage (ac relays)	3.14	4.9.11		
Current range (ac relays)	3.15	4.9.12		
Zero crossover (ac relays)	3.16	4.9.13		
Bias current (when applicable)	3.17	4.9.14		
Maximum deactuation voltage	3.18	4.9.15		
All relay classes				
Seal	3.7	4.9.4		
Visual and mechanical inspection (external) (dimensional check on 2 sample units only)	3.1, 3.3, 3.4, 3.39 and 3.40	4.9.1		
Group II				
All relay classes				
Transients	3.19	4.9.16	(See next page for Group II sample units and failures)	
Timing cycle	3.20	4.9.17		
Recycle time	3.21	4.9.18		
Reverse polarity (dc operated relays)	3.22	4.9.19		
Insulation resistance	3.8	4.9.3		
Dielectric withstanding voltage	3.9	4.9.5		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.10.1	4.9.7.1		
Pickup time and dropout time	3.10.2	4.9.7.2		
Contact dynamic characteristics	3.10.3	4.9.7.3		
Input current (when applicable)	3.11	4.9.8		

See footnotes at end of table.

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TABLE VI. Qualification inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Group II - Continued				
The following tests are applicable to class C relays only				
Leakage current	3.12	4.9.9	4	1
Waveform distortion (ac relays)	3.13	4.9.10		
DC offset voltage (ac relays)	3.14	4.9.11		
Current range (ac relays)	3.15	4.9.12		
Zero crossover (ac relays)	3.16	4.9.13		
Bias current (when applicable)	3.17	4.9.14		
Maximum deactuation voltage	3.18	4.9.15		
All relay classes				
Seal	3.7	4.9.4		
Visual and mechanical inspection (external)	3.1,3.3,3.4, 3.39 and 3.40	4.9.1		
Group III				
All relay classes				
Electromagnetic interference	3.23	4.9.20	4	1
Crosstalk (when specified)	3.24	4.9.21		
Capacitance (when specified)	3.25	4.9.22		
Continuous current	3.26	4.9.23		
Resistance to soldering heat (not applicable to socket plug-in relay)	3.27	4.9.24		
Salt spray (corrosion)	3.28	4.9.25		
Insulation resistance	3.8	4.9.5		
Dielectric withstanding voltage	3.9	4.9.6		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.10.1	4.9.7.1		
Pickup time and dropout time	3.10.2	4.9.7.2		
Contact dynamic characteristics	3.10.3	4.9.7.3		
Input current (when applicable)	3.11	4.9.8		
The following tests are applicable to class C relays only				
Leakage current	3.12	4.9.9		
Waveform distortion (ac relays)	3.13	4.9.10		
DC offset voltage (ac relays)	3.14	4.9.11		
Current range (ac relays)	3.15	4.9.12		
Zero crossover (ac relays)	3.16	4.9.13		
Bias current (when applicable)	3.17	4.9.14		
Maximum deactuation voltage	3.18	4.9.15		

See footnotes at end of table.

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TABLE VI. Qualification inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Group III - Continued				
All relay classes				
Seal	3.7	4.9.4	(See previous page for Group III sample units and failures)	
Visual and mechanical inspection (external)	3.1,3.3,3.4, 3.39 and 3.40	4.9.1		
Group IV				
All relay classes				
Thermal shock	3.29	4.9.26		
Low temperature operation	3.30	4.9.27		
Shock (specified pulse)	3.31	4.9.28		
Vibration	3.32	4.9.29		
Accleration	3.33	4.9.30		
Resistance to solvents	3.34	4.9.31		
Strength of terminals and mounting studs 2/	3.35	4.9.32		
Insulation resistance	3.8	4.9.5		
Dielectric withstanding voltage	3.9	4.9.6		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.10.1	4.9.7.1		
Pickup time and dropout time	3.10.2	4.9.7.2		
Contact dynamic characteristics	3.10.3	4.9.7.3		
Input current (when applicable)	3.11	4.9.8	4	1
The following tests are applicable to class C relays only				
Leakage current	3.12	4.9.9		
Waveform distortion (ac relays)	3.13	4.9.10		
DC offset voltage (ac relays)	3.14	4.9.11		
Current range (ac relays)	3.15	4.9.12		
Zero crossover (ac relays)	3.16	4.9.13		
Bias current (when applicable)	3.17	4.9.14		
Maximum deactuation voltage	3.18	4.9.15		
All relay classes				
Seal	3.7	4.9.4		
Visual and mechanical inspection (external)	3.1,3.3,3.4, 3.39 and 3.40	4.9.1		

See footnotes at end of table.

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TABLE VI. Qualification inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Group V				
All relay classes				
Overload	3.36	4.9.33	For qualification to level "L"	
Life 3/	3.37	4.9.34		
Insulation resistance	3.8	4.9.5	6	0
Dielectric withstanding voltage	3.9	4.9.6	12	1
Electrical characteristics:			22	2
Static contact resistance (or voltage drop)	3.10.1	4.9.7.1	38	5
Pickup time and dropout time	3.10.2	4.9.7.2		
Contact dynamic characteristics	3.10.3	4.9.7.3	For qualification to level "M"	
	3.11	4.9.8	22	0
			38	1
The following tests are applicable to class C relays only				
Leakage current	3.12	4.9.9		
Waveform distortion (ac relays)	3.13	4.9.10		
DC offset voltage (ac relays)	3.14	4.9.11		
Current range (ac relays)	3.15	4.9.12		
Zero crossover (ac relays)	3.16	4.9.13		
Bias current (when applicable)	3.17	4.9.14		
Maximum deactuation voltage	3.18	4.9.15		
All relay classes				
Seal 2/	3.7	4.9.4		
Visual and mechanical inspection (external)	3.1, 3.3, 3.4, 3.39 and 3.40	4.9.1		
Group VI				
Intermediate current 4/	3.38	4.9.35	4	0
Insulation resistance	3.8	4.9.5		
Dielectric withstanding voltage	3.9	4.9.6		
Electrical characteristics:				
Static contact resistance (or voltage drop)	3.10.1	4.9.7.1		
Pickup time and dropout time	3.10.2	4.9.7.2		
Contact dynamic characteristics	3.10.3	4.9.7.3		
Input current (when applicable)	3.11	4.9.8		
The following tests are applicable to class C relays only				
Leakage current	3.12	4.9.9		
Waveform distortion (ac relays)	3.13	4.9.10		
DC offset voltage (ac relays)	3.14	4.9.11		
Current range (ac relays)	3.15	4.9.12		

See footnotes at end of table.

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TABLE VI. Qualification inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units to be inspected	Number of failures allowed
Group VI - Continued				
Zero crossover (ac relays)	3.16	4.9.13	(See previous page for Group VI sample units and failures)	
Bias current (when applicable)	3.17	4.9.14		
Maximum deactuation voltage	3.18	4.9.15		
All relay classes				
Seal	3.7	4.9.4		
Visual and mechanical inspection (external)	3.1,3.3,3.4, 3.39 and 3.40	4.9.1		

1/ One unsealed sample unit shall be subjected to the internal inspection, and when applicable, this same unsealed sample unit shall be subjected to coil inductance test.

2/ Test four sample relays subjected to the highest current life test. These four samples shall be subjected to terminal strength and seal. No failures shall be allowed.

3/ The sample size shall be equally divided among the specified contact rating and shall be of sufficient size to test a minimum of two relays per contact rating with rated loads on all contacts.

4/ Intermediate current operation test cycles apply to life failure rate qualification.

TABLE VII. Requirements for extension of qualification to lower (better) failure rate levels (90 percent confidence level).

Failure rate level symbol	Qualified failure rate level (%/10,000 operations)	Cumulative unit operations in millions					
		C 1/ = 0	C = 1	C = 2	C = 3	C = 4	C = 5
M	1.0	2.30	3.89	5.32	6.68	7.99	9.27
P	0.1	23.0	38.9	53.2	66.8	79.9	92.7
R	0.01	230	389	532	668	799	927

1/ C = Acceptance number or number of failures permitted.

TABLE VIII. Sampling plan for maintenance of FR level qualification (60 percent confidence level).

FR level symbol	Qualified FR (%/10,000 operations)	Maximum qualification maintenance period	Cumulative unit operations in millions					
			C 1/ = 0	C = 1	C = 2	C = 3	C = 4	C = 5
L	3.0	6 months	.306	.673	1.03	1.39	1.75	2.10
M	1.0	6 months	.916	2.02	3.10	4.18	5.25	6.30
P	0.1	12 months	9.15	20.2	31.0	41.8	52.5	63.0
R	0.01	24 months	91.6	202	310	418	525	630

1/ C = Acceptance number or number of failures permitted.

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4.7 FR level determination. Determination of FR levels shall be based upon data from all completed life and intermediate current tests. Data for determination of FR levels shall be accumulated in the following manner:

a. The qualification test samples.

b. Test on every production lot which has been submitted for groups B and C quality conformance inspection.

c. Test results of relays which have a similar internal design and construction, same production processes, same or higher environmental capability and same electrical characteristics as the qualified relay and which have been tested in accordance with the groups B and C quality conformance inspection conditions. These relay types must be reviewed and approved by the activity responsible for qualification prior to being considered as acceptable relays for use in obtaining FR data.

4.7.1 Records. Test records shall be maintained which shall include the data derived from the sources specified in 4.7. The example forms shown in MIL-STD-690 include the minimum information required and a suggested format for this report. Test data on every production lot that has been submitted for quality conformance inspection shall be included.

4.7.2 FR level computation. Data from test records shall be used to compute the observed failure rates of the sample units using a maintenance of FR record forms similar to figure 1 of MIL-STD-690. For this computation, a sample unit which fails at any given operation shall be classed as a failure for all subsequent measurements. However, the contractor, at his option, may physically remove failed sample units from the test. If a failed sample unit is not removed, the operations accmulated subsequent to its failure shall not be counted toward the cumulative component operations shown on record form.

4.7.2.1 Contact FR computation. The data from tests shall be recorded in such a manner so that, if necessary, it can be analyzed to compute contact FR.

4.7.3 Exemption of data. Exemptions of data requirement in MIL-STD-690 shall apply.

4.7.4 Qualification approval for higher FR. Qualification approval granted for one of the lower FR levels will include qualification for all of the higher FR levels, e.g., qualification approval for level R will include qualification approval for levels P, M, and L.

4.8 Quality conformance inspection.

4.8.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection.

4.8.1.1 Inspection lot. An inspection lot shall consist of all relays covered by a single specification sheet, produced and sealed under essentially the same conditions, and offered for inspection at one time within a period not to exceed 1 month.

4.8.1.2 Group A inspection. Group A inspection shall consist of the inspections specified in table IX, in the order shown.

4.8.1.2.1 Sampling plan. The tests in subgroups 2 and 3 shall be perfomred on each relay offered for inspection except as noted. For subgroup 1, sampling shall be as specified in table IX. The test for internal moisture for failure rate level L shall be performed on each inspection lot by statistical sampling and inspection in accordance with MIL-STD-105 for normal inspection. The degree of sampling (normal or tightened) shall be determined independently for each lot as applicable. If, during the 100 percent inspection of subgroup 2 over 5 percent of the relays are discarded, the lot shall be rejected. When possible, these tests shall be witnessed by the government inspector at the time they are performed by the contractor.

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TABLE IX. Group A inspection.

Inspection	Requirement paragraph	Test method paragraph	AQL (percent defective
Subgroup 1			
Level L only			
Screening, internal moisture -	3.5	4.9.2.1	AQL percentage defective major 4.0
Screening- - - - - - - - - - -	3.5	4.9.2	100% inspection discard all failed relays
Level M, P or R			
Vibration (sinusoidal) - - - -	3.32	4.9.29	100% inspection discard all failed relays
Screening, internal moisture -	3.5	4.9.2	
Subgroup 2 1/			
Insulation resistance- - - - - -	3.8	4.9.5	
Dielectric withstanding voltage-	3.9	4.9.6	
Electrical characteristics:			
Static contact resistance (or voltage drop)- - - - - - - - -	3.10.1	4.9.7.1	
Pickup time and dropout time -	3.10.2	4.9.7.2	
Contact dynamic characteristics	3.10.3	4.9.7.3	
Input current (when applicable)-	3.11	4.9.8	100% inspection discard all failed relays
The following tests are applicable to class C relays only			
Leakage current- - - - - - - - -	3.12	4.9.9	
Waveform distortion (ac relays)-	3.13	4.9.10	
DC offset voltage (ac relays)- -	3.14	4.9.11	
Current range (ac relays)- - - -	3.15	4.9.12	
Zero crossover (ac relays) - - -	3.16	4.9.13	
Bias current (when applicable) -	3.17	4.9.14	
Maximum deactuation voltage- - -	3.18	4.9.15	
Subgroup 3			
Solderability- - - - - - - - - -	3.6	4.9.3	2/
Seal - - - - - - - - - - - - - -	3.7	4.9.4	100% inspection discard all failed relays
Visual and mechanical inspection (external) - - - - - - - - - -	3.1,3.3,3.4, 3.39 and 3.40	4.9.1	100% inspection discard all failed relays 4/

1/ Testing sequence optional.
2/ Solderability shall be required on two sample relays drawn at random from each inspection lot. In the event of a failure, the entire lot shall be reworked and a second random sample taken. When rework requires the replating of the relay, the entire lot shall be resubmitted for all group A inspection after passing solderability. If the rework requires only the solder dipping of the terminals, then the entire lot shall be resubmitted for subgroups 2 and 3 testing, after passing solderability.
3/ Physical dimensions and weight shall be measured on two sample units per lot.
4/ Minor defects, such as marking, may be reworked.

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4.8.1.2.2 Rejected lots.

a. Subgroup 1, level L. If an inspection lot is rejected by the applicable statistical sampling, the manufacturer shall either discard the entire lot or test the enitre lot to subgroup 1 test. Subgroup 1 test may be performed either before or after subgroup 2 at the option of the manufacturer, but must be performed on all relays from a lot which has been rejected due to failure to meet subgroup test.

b. Subgroup 2. Rejected lots shall not be offered for reinspection.

c. Subgroup 3. See footnotes 2/, 3/, and 4/ of table IX.

4.8.1.2.3 Manufacturer's production inspection. If the manufacturer performs tests similar to those specified in subgroup 3, table IX, as the final step of his production process, group A, subgroup 3, inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 2 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

a. Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than that specified for subgroup 3. Test conditions shall be equal to or more stringent than those specified for subgroup 3.

b. Manufacturer subjects 100 percent of the product supplied under this specification to his production tests.

c. The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

d. The lot rejection criterion is the same or more stringent than that specified herein.

e. The manufacturer shall make available all information concerning the test procedures and instrumentation used in his production tests. This data shall be provided as part of the evaluation required for MIL-STD-790. The manufacturer shall also make available to the Government all records of all detail test data resulting from production tests.

4.8.2 Periodic inspection. Periodic inspection shall consist of groups B and C. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.8.2.2.4), delivery of products which have passed group A shall not be delayed pending the results of these periodic inspections.

4.8.2.1 Group B inspection. Group B inspection shall consist of the test specified in table X and shall be made on sample units which have been subjected to and have passed the group A inspection.

TABLE X. Group B inspection.

Inspection	Requirement paragraph	Test method paragraph
Life 1/	3.37	4.9.33
Insulation resistance	3.8	4.9.3
Dielectric withstanding voltage	3.9	4.9.5
Electrical characteristics:		
Static contact resistance (or voltage drop)	3.10.1	4.9.7.1
Pickup time and dropout time	3.10.2	4.9.7.2
Contact dynamic characteristics	3.10.3	4.9.7.3

See footnote at end of table.

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TABLE X. Group B inspection. - Continued.

Inspection	Requirement paragraph	Test method paragraph
Input current (when applicable)- - - - - -	3.11	4.9.8
The following tests are applicable to class C relays only		
Leakage current- - - - - - - - - - - - - - -	3.12	4.9.9
Waveform distortion (ac relays)- - - - - -	3.13	4.9.10
DC offset voltage (ac relays)- - - - - - -	3.14	4.9.11
Current range (ac relays)- - - - - - - - -	3.15	4.9.12
Zero crossover (ac relays) - - - - - - - -	3.16	4.9.13
Bias current (when applicable) - - - - - -	3.17	4.9.14
Maximum deactuation voltage- - - - - - - -	3.18	4.9.15
All relay classes		
Seal -	3.7	4.9.4
Visual and mechanical inspection (external) - - - - - - - - - - - - - - - -	3.1,3.3,3.4, 3.39 and 3.40	4.9.1

1/ Sample units shall be tested in the following sequency of contact loads, as applicable: First, resistive; second, inductive; third, low level; others, optional. Each sample shall be tested at only one contact rating with rated loads on all contacts.

4.8.2.1.1 Sampling plan. The test sample size shall be determined by the manufacturer so that the relay operations generated meet the maintenance of qualification requirements specified for the qualified failure rate level (see 4.5.4). In any event, a minimum of one sample unit shall be selected from each lot. As far as is practicable, the relays tested during a maintenance period shall be representative of all relays produced during this period. The accumulated data shall be used for maintenance and extension of failure rate qualification. Data from life test sample units and intermediate current sample units from group C inspection may be included in this sample.

4.8.2.1.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or purchase order.

4.8.2.2 Group C inspection. Group C inspection shall consist of the tests specified in table XI. Group C inspection shall be made on sample units selected from inspection lots which have passed the group A inspection.

TABLE XI. Group C inspection.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units for inspection
Group I (every month)			
Intermediate current - - - - - -	3.38	4.9.35	
Strength of terminals and mounting studs - - - - - - - -	3.35	4.9.32	
Insulation resistance- - - - - -	3.8	4.9.5	(See next page for Group I sample units)
Dielectric withstanding voltage-	3.9	4.9.6	
Electrical characteristics:			
Static contact resistance (or voltage drop)- - - - - - - - -	3.10.1	4.9.7.1	
Pickup time and dropout time -	3.10.2	4.9.7.2	
Contact dynamic characteristics - - - - - - - - - - - - - -	3.10.3	4.9.7.3	

See footnote at end of table.

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TABLE XI. Group C inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units for inspection
Group I (every month) - Cont.			
Input current (when applicable)-	3.11	4.9.8	3
The following tests are applicable to class C relays only			
Leakage current-	3.12	4.9.9	
Waveform distortion (ac relays)-	3.13	4.9.10	
DC offset voltage (ac relays)-	3.14	4.9.11	
Current range (ac relays)-	3.15	4.9.12	
Zero crossover (ac relays)	3.16	4.9.13	
Bias current (when applicable)	3.17	4.9.14	
Maximum deactuation voltage-	3.18	4.9.15	
All relay classes			
Seal	3.7	4.9.4	
Visual and mechanical inspection (external)	3.1,3.3,3.4, 3.39 and 3.40	4.9.1	
Group II (every 6 months)			
Timing cycle	3.20	4.9.17	2
Recycle time	3.21	4.9.18	
Thermal shock 1/	3.29	4.9.26	
Shock (specified pulses)	3.31	4.9.28	
Vibration 1/	3.32	4.9.29	
Strength of terminals and mounting studs	3.35	4.9.32	
Insulation resistance-	3.8	4.9.5	
Dielectric withstanding voltage-	3.9	4.9.6	
Electrical characteristics:			
Static contact resistance (or voltage drop)-	3.10.1	4.9.7.1	
Pickup time and dropout time -	3.10.2	4.9.7.2	
Contact dynamic characteris tics	3.10.3	4.9.7.3	
Input current (when applicable)	3.11	4.9.8	
The following tests are applicable to class C relays only			
Leakage current-	3.12	4.9.9	
Waveform distortion (ac relays)-	3.13	4.9.10	
DC offset voltage (ac relays)-	3.14	4.9.11	
Current range (ac relays)-	3.15	4.9.12	
Zero crossover (ac relays)	3.16	4.9.13	
Bias current (when applicable)	3.17	4.9.14	
Maximum deactuation voltage-	3.18	4.9.15	
All relay classes			
Seal	3.7	4.9.4	
Visual and mechanical inspection (external)	3.1,3.3,3.4, 3.39 and 3.40	4.9.1	

See footnote at end of table.

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TABLE XI. Group C inspection - Continued.

Inspection	Requirement paragraph	Test method paragraph	Number of sample units for inspection
Group III (every 6 months)			
Overload (highest resistive load)	3.36	4.9.33	2
Life (highest resistive load)	3.37	4.9.34	
Strength of terminals and mounting studs	3.35	4.9.32	
Insulation resistance	3.8	4.9.5	
Dielectric withstanding voltage	3.9	4.9.6	
Electrical characteristics:			
Static contact resistance (or voltage drop)	3.10.1	4.9.7.1	
Pickup time and dropout time	3.10.2	4.9.7.2	
Contact dynamic characteristics	3.10.3	4.9.7.3	
Input current (when applicable)	3.11	4.9.8	
The following tests are applicable to class C relays only			
Leakage current	3.12	4.9.9	
Waveform distortion (ac relays)	3.13	4.9.10	
DC offset voltage (ac relays)	3.14	4.9.11	
Current range (ac relays)	3.15	4.9.12	
Zero crossover (ac relays)	3.16	4.9.13	
Bias current (when applicable)	3.17	4.9.14	
Maximum deactuation voltage	3.18	4.9.15	
All relay classes			
Seal	3.7	4.9.4	
Visual and mechanical inspection (external)	3.1,3.3,3.4, 3.39 and 3.40	4.9.1	

1/ Written reports of the results of insulation resistance are required only as a part of thermal shock; written reports of the results of contact resistance and dielectric withstanding voltages are required only as a part of vibration.

4.8.2.2.3 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.8.2.2.4 Noncompliance. If a sample fails to pass the group B or C inspection, the manufacturer shall notify the qualifying acitivty and the cognizant inspection activity of such failure and take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same materials and processes, and which are considered subject to the same failure. Acceptance and shipment of the product shall be discontinued until corrective action, acceptable to the qualifying activity has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the qualifying activity). Groups A and B inspection may be reinstituted; however, final acceptance and shipment shall be withheld until the group C inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure shall be furnished to the cognizant inspection activity and the qualifying activity.

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4.8.3 Inspection of packaging. The sampling and inspection of the preservation and interior pack marking shall be in accordance with the group A and group B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.9 Methods of inspection.

NOTE: As used herein, "coil" on class A and class B relays may be used interchangeably for "input" on class C relays; and "contacts" on class A and class B relays may be used interchangeably for "output" on class C relays.

4.9.1 Visual and mechanical. Relays shall be inspected to verify that the materials, external design and construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.1, 3.3, 3.4, 3.39 and 3.40).

4.9.2 Screening (see 3.5).

4.9.2.1 Internal moisture (applicable to classes A and B). Relays (coils deenergized) shall be held at 15°C to 25°C for a minimum of 6 hours. The insulation resistance of all contact pins to case only, shall be measured and observed. The relay coil shall be energized with 140 percent of maximum input voltage for a period of 2-1/2 minutes. The insulation resistance of all contact pins to case only shall be verified a minimum of once each 30 seconds during this period and the lowest value shall meet the requirements of 3.5.

4.9.2.2 High temperature storage (stablization bake). Relays shall be tested in accordance with method 1008 of MIL-STD-883. Unless otherwise specified, test condition B and a time duration of 24 hours minimum shall apply.

4.9.2.3 Temperature cycling. Relays shall be tested in accordance with method 1010 of MIL-STD-883, test condition B. The following details shall apply:

a. Special mounting: Relays shall be suspended in the test chamber by normal mounting means in a plane parallel to the normal air flow. Test leads may be used for mounting; however, they shall not provide a heat sink.

b. End-point measurements and inspections: For class A and class B relays, run-in (see 4.9.2.4) may be performed concurrent with or subsequent to this test. For class C relays, the dielectric withstanding voltage shall be as specified in 4.9.6.

4.9.2.4 Run-in (applicable to classes A and B).

4.9.2.4.1 Failure rate level L. Relays shall be subjected to a 5,000-cycles run-in test. The cycling rate shall be a maximum of:

$$\frac{.1}{\text{maximum operate time (seconds)} + \text{maximum release time (seconds)}} \text{ cycles}$$

per second where the operate and release times are those of the relay under test. The coil shall be energized at maximum input voltage or current. Relays shall have the contacts loaded as follows: Open circuit load voltage 10 to 50 mV (dc or peack ac). The load current shall be 10 to 50 µA. Each pair of mated contacts shall be monitored during each operation in accordance with 4.9.2.4.1.1.

4.9.2.4.1.1 Monitoring. The contact resistance or voltage drop, as applicable, shall be monitored during 40 percent minimum of each "on" and each "off" period. The monitoring equipment shall automatically turn off when a failure occurs, or shall record every failure.

4.9.2.4.2 Failure rate levels M, P, and R.

a. High temperature. Qualification inspection only; relays shall be subjected to 125°C with maximum input voltage or current on the coil for 1 hour. For group A testing, relays shall be subjected to 125°C; the test chamber shall

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stabilize at 125°C after the specimens have been inserted into the test chamber. The test shall be performed with maximum input voltage or current on the coil for 15 minutes minimum for relays weighing one ounce or less; 30 minutes minimum for relays weighing over one ounce but not more than .3 pound; and 1 hour minimum for relays weighing .3 pound to 3 pounds maximum. The pickup voltage shall be measured to determine compliance with 3.1. While at this temperature, the relays shall be subjected to a 2,500 cycle run-in test. The operation rate shall be in accordance with the rate outlined in 4.9.2.4.1. Relays shall have the contacts loaded as follows: Open circuit load voltage 10 to 50 mV dc or peak ac. The load current shall be 10 to 50 μA. Each pair of mated contacts shall be monitored during each operation in accordance with 4.9.2.4.1.1.

b. Low temperature. Following high temperature, for qualification only, relays shall be subjected to -65°C with the coil or coils deenergized for 1 hour. For group A testing, relays shall be subjected to -65°C; the test chamber shall stabilize at -65°C after the specimens have been inserted into the test chamber. The test shall be performed with the coil or coils deenergized for fifteen minutes minimum for relays weighing one ounce or less; minutes minimum for relays weighing over one ounce but not more than .3 pound; and 1 hour minimum for relays weighing .3 pound ro 3 pounds maximum. At the end of either period, the dropout voltage shall be measured. While at this temperature, the relays shall be subjected to a 2,500 cycle run-in test in accordance with the procedure outlined in 4.9.2.4.2a.

4.9.2.5 Load conditioning (applicable to class C only). Not to exceed specified maximum limits, relays shall be cycled by applying a step function voltage to the input; the input shall be energized at zero or rated input voltage. The load shall be maximum rated current, without auxiliary heat sink, and each output circuit shall be loaded with the maximum rated resistive current at the highest rated voltage, for 3 hours, at a rate no less than 1 nor more than 30 operations per second. When applicable (see 3.1), the bias shall be applied at the rated value. The relay shall be turned off 10 percent of the time and turned on 90 percent of the time. For SPDT and DPDT relays, cycling shall be 50 percent turned on and 50 percent turned off; or for 1.5 hours of test, 10 percent on and 90 percent off, and 90 percent on and 10 percent off for 1.5 hours of test.

4.9.2.6 Burn-in test. Relays shall be tested in accordance with method 1015 of MIL-STD-883. The following details and exceptions shall apply:

a. Test duration of 160 hours for qualification and 24 hours for group A testing.

b. Test loads: Classes A and B as specified in 4.9.2.4.1
Class C as specified in 4.9.2.5

c. Test temperature: 125°C minimum ambient.

d. Test mounting: As specified in 4.9.2.3a.

e. Not applicable.

f. Not applicable.

g. Not applicable.

4.9.3 Solderability (see 3.6). Relays shall be tested in accordance with method 208 of MIL-STD-202. The following detail and exception shall apply:

a. Number of terminations of each part to be tested: All.

b. The temperature of the molten solder shall be a uniform 260°C ±5°C (500°F ±9°F).

4.9.4 Seal (see 3.7). Relays shall be tested in accordance with 4.9.4.1 or 4.9.4.2 as applicable.

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4.9.4.1 Hermetically-sealed (potted) relays (see 3.7.1.1). Relays shall be tested as follows:

a. Prior to installing a gasket on the relay header (if a gasket is used), the relay shall be totally immersed in a container of "tap" water. (The term "tap" water as used here means ordinary drinking water that has not been altered in any way, such as by the addition of any other substance, distilling, etc.). The part of the relay closest to the surface of the water shall be a minimum of 1 inch (25.4 mm) below this surface.

b. The container and water-covered relay shall then be placed in a vacuum chamber. The chamber shall be sealed. The chamber pressure shall be reduced from room ambient to 1 inch (75,000 feet) +0.0 -0.2 inch of mercury within 5 minutes, and shall be maintained at this level for 30 minutes minimum. The chamber pressure shall be increased to room ambient within 1 minute, and shall be maintained at room ambient pressure for 30 minutes, minimum. The foregoing shall constitute one cycle. The relay shall remain fully immersed in the water during the cycle.

c. Within a maximum of one-half hour after the cycle, each relay shall be removed from the water and dried by shaking, wiping or blowing with contaminant-free air or gas, but not by any form of heating or baking.

d. Within a maximum of one-half hour after drying, each relay shall pass the insulation resistance and dielectric withstanding voltage tests as specified in 4.9.5 and 4.9.6.

4.9.5 Insulation resistance (see 3.8). Relays shall be tested in accordance with method 302 of MIL-STD-202 with the relay in the energized and deenergized positions. The following details shall apply:

a. Test condition A (for relays with coil and contact ratings both below 60 volts), and B (for all other relays).

b. Points of measurements: As specified in points of application in table XII.

4.9.6 Dielectric withstanding voltage (see 3.9). Relays shall be tested as specified in 4.9.6.1 and, when specified (see 3.1), in accordance with 4.9.6.2. Testing in accordance with 4.8.9.2 is not required for group A testing.

4.9.6.1 At atmospheric pressure. Relays shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

a. Points of application and magnitude of test voltage: As shown in table XII.

b. Maximum leakage current: 100 μA.

c. Duration of application: 60 seconds for qualification and groups B and C tests; 5 seconds for group A tests.

TABLE XII. Test details for dielectric withstanding voltage.

Points of application	Test voltage
Between case, frame, or enclosure, and between all contacts in the energized and deenergized position Between case, frame, or enclosure and input leads Between all contacts and input leads Between open contacts in the energized and deenergized position Between coils of dual-coil relays Between contact poles in the energized and deenergized positions	1,000 volts ac plus twice rated voltage or as specified (see 3.1)

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4.9.6.2 At reduced barometric pressure. Relays specified for operation above 10,000 feet shall be tested in accordance with method 105 of MIL-STD-202. The following details shall apply:

a. Method of mounting: Normal mounting means.

b. Test condition C.

c. Tests during subjection to reduced pressure: As specified in 4.9.6.1, except test voltage shall be 350 volts.

Following these tests, relays shall be inspected for evidence of arcing, flashover, insulation breakdown, and damage.

4.9.7 Electrical characteristics.

4.9.7.1 Static contact resistance (or voltage drop) (see 3.10.1).

4.9.7.1.1 Static Contact resistance (class A relays). Relays shall be tested in accordance with method 307 of MIL-STD-202. The following details and exception shall apply:

a. Method of connection: For relays with wire-lead terminals, this measurement shall be made 1/8-inch maximum from the emergence of the lead from the relay.

b. Test current and voltage: Contacts shall be loaded with 30 mV maximum (dc or peak ac) at 10 mA maximum.

c. Test current and voltage after high level life tests: Contacts shall be loaded up to rated current and rated voltage.

d. Points of measurement: All mated contacts in their closed position; the coil shall be energized with rated voltage if necessary to effect contact closure.

e. Number of activations prior to measurement: None.

f. Number of test activations: Three (no contact voltage shall be applied during contact transfer).

g. Number of measurements per activation: One in each closed contact position.

4.9.7.1.2 Contact voltage drop (class B relays). The contact voltage drop across the relay contacts shall be measured at the points to which external circuits are normally connected. Normally closed and normally open contacts shall be measured separately. Relay contacts rated over 2 amperes shall carry rated resistive current at 6 V dc except Group A testing of relays rated for low level to high level loads the contacts shall be loaded as specified in 4.9.7.1.1b. When specified after the load tests, the contact voltage drop shall be measured at the terminals corresponding to the tested contacts. The contacts shall be caused to break and then make the test current before each of 10 consecutive measurements. Coil voltage shall be as specified. Individual readings and the average value of 10 consecutive readings shall meet the requirements of 3.10.1 and not exceed the values specified in table IV. In performing the contact voltage drop tests on plug-in relays, and in the event of a reading that exceeds the maximum allowable contact voltage drop when measured external to the connector, a measurement may be made directly at the pins of the relay. If the readings are then within the allowable limits, the relay will be considered to have passed. In the case of relays with potted connections, special provisions should be made to allow for voltage drop due to resistance of leadwires. For group A, one reading per contact will suffice and may be performed at a lower current level. At the option of the manufacturer, the relay contacts may be closed prior to the application of test current. The contact flow of current begins. The contact voltage drop shall be within the average limits specified in table IV adjusted for the lower current level used in test.

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4.9.7.1.3 Class C relays (output voltage drop). With the relay connected to nominal system voltage and rated load connected, operate relay normally, and check voltage difference between input and output to connected operating load.

4.9.7.2 Pickup time and dropout time (applicable to class A and B relays) (see 3.10.2). The pickup time and dropout time shall be measured using an electronic clock or digital oscilloscope. The operating mode of the relay shall be as specified (see 3.1). The pickup time and dropout time shall be exclusive of contact bounce and timing measurements shall be made on all contacts.

4.9.7.3 Contact dynamic characteristics (classes A and B only) (see 3.10.2).

4.9.7.3.1 Contact bounce (applicable to failure leve rate L) (see 3.10.2.1). Contact bounce shall be measured on each contact set using an oscilloscope or the acceptable means approved by the qualifying activity. The trace shall show contact switching at operate and release and appropriate timing markers. Maximum input voltage shall be applied. Contacts shall be loaded with 6 volts dc maximum or peak ac at 10 mA maximum. After high level rated load life and intermediate current tests, contact bounce shall be measured with contact loaded at test current and test voltage. A contact bounce shall be considered any occurrence equal to or greater than 90 percent of the open circuit voltage with a pulse width of 10 microseconds or greater. The circuit shown on figure 2, or equivalent, shall be used.

4.9.7.3.2 Contact stabilization time (applicable to failure rate levels M, P and R) (see 3.10.2.2). Contact stabilization time shall be measured on each contact set using an oscilloscope or other acceptable means approved by the qualifying activity. The equipment shall have the capability of indicating a failure pulse width of greater than 1 microsecond. The trace shall show contact switching at operate and release, appropriate timing markers and maximum allowable contact voltage drop markers (which equates to the allowable static contact resistance). Maximum input voltage shall be applied. Contacts shall be loaded with 50 mV dc maximum or peak ac at 50 mA maximum. Contact stabilization time shall be defined as the maximum time allowed for the contacts to reach and maintain a static contact resistance state following the actual operate or release time of the relay (essentially, it is the sum of the contact bounce time and the time for the dynamic contact resistance to stabilize to the static contact resistance). After high level rated load life and intermediate current tests, the contacts shall be loaded at the test current and test voltage and contact bounce in lieu of contact stabilization time shall be measured as specified in 4.9.7.4.1.

a. The switching relay shall be a mercury wetted contact switching relay. E is the maximum input voltage for the relay under test. The switching relay is operated from a source voltage independent of E. CRO is cathode-ray oscilloscope with a rise time of .020 microseconds or less. The horizontal (time) deflection scale shall be set at 1/2 to 1 millisecond per division (1/2 to 1 MS/CM), and the vertical (voltage) defection to be such that the vertical gain provides accuracy in reading. Example: 5 V/CM for a 12 V BEMF limit, or 10 V/CM for a 24 V BEMF limit. Both horizontal (time) and vertical (voltage) deflection traces shall be of a calibrated grid spacing (i.e., CM or MM).

b. The source, E, shall be a low impedance source (such as batteries) capable of delivering the maximum input voltage with no limiting resistor or potentiometer used to regulate the line voltage. (Having low source impedance representing a typical application is more important than having the exact maximum input voltage.)

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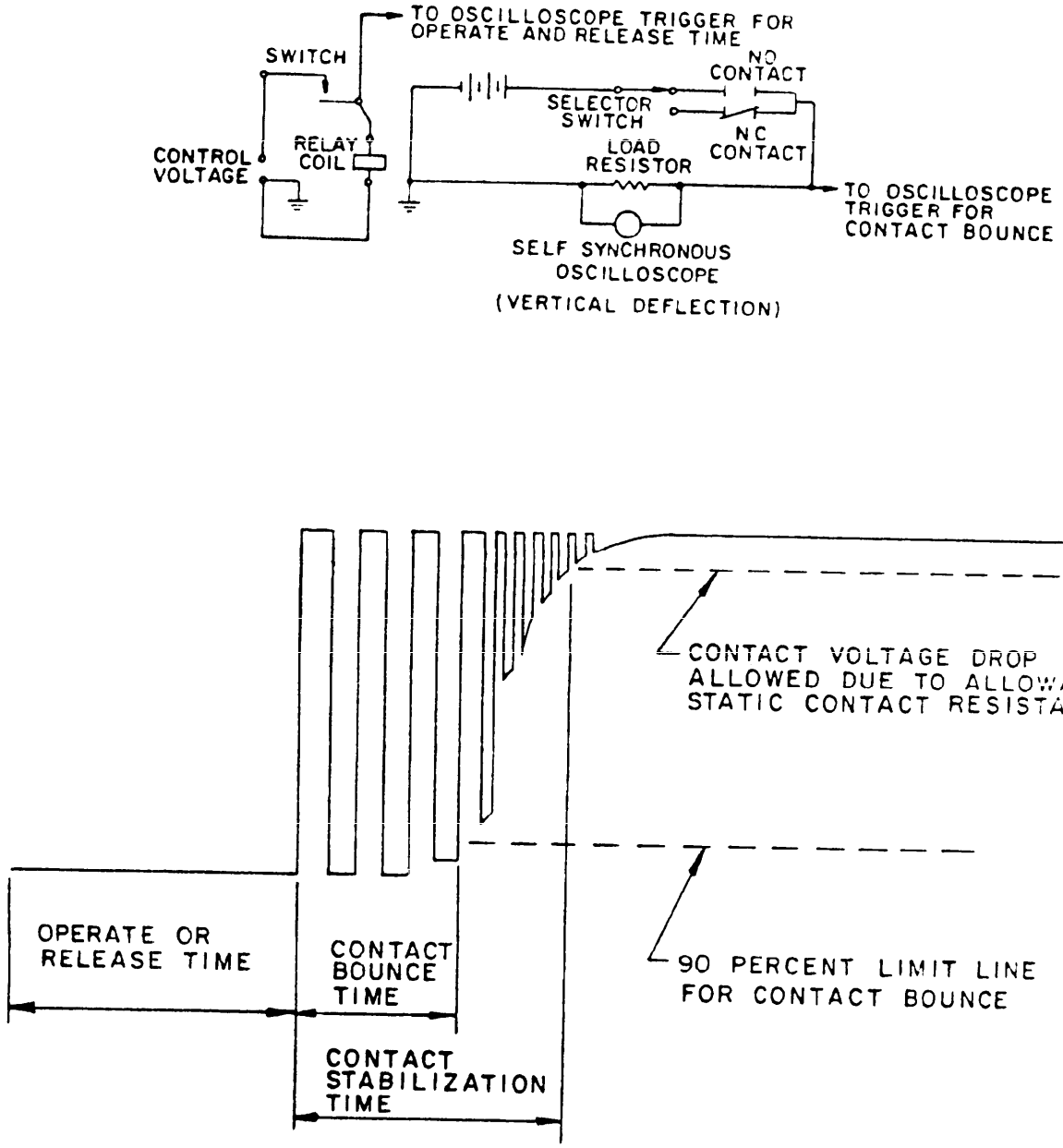


FIGURE 2. Typical circuit for operate and release time, contact bounce and contact stabilization time with typical traces.

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c. The switching relay shall be closed for a minimum of ten times the operate time of the test relay, or switching relay whichever is longer, to allow the oscilloscope and circuit network to stabilize and then opened to obtain the inducted voltage deflection trace. Relay to be driven at a 10 ±2 Hz cycling rate with approximatley equal open and closed times.

(1) The reading shall be observed on the oscilloscope. The magnitude of the induced voltage transient shall be noted. A typical trace is presented on figure 4.

(2) The maximum value of three consecutive readings shall be noted.

(3) Unless otherwise specified, the temperature at the time of the testing shall be 25°C ±5°C.

d. For group A testing, only one measurement and other suitable test methods may be used to verify that the back electromotive force (EMF) (coil kick) is within the specified limit.

4.9.8 Input current (3.1). The maximum operating voltage (see 3.1) shall be applied to the input and control terminals, and the input current and, if applicable, the control current shall be measured during the timing cycle and during steady-state.

4.9.9. Leakage current (class C relays) (see 3.12). With the relay connected to maximum system voltage and rated load connected, operate relay normally. During the load off interval, measure the leakage current to the load.

4.9.10 Waveform distortior (class C, ac relays) (see 3.13).

a. Apply the rated voltages (see 3.1) to the input and bias (when applicable). Apply rated output voltage and adjust the load to obtain rated current with relay turn on.

b. Connect an oscilloscope across relay output and adjust oscilloscope to display one complete cycle of relay output voltage.

c. With relay turned on, observe output voltage on oscilloscope. Output voltage distortion shall not exceed the limits specified (see 3.1).

4.9.11 DC offset voltage (class C, ac relays) (see 3.14).

a. Apply rated voltages and currents to the output and bias (when applicable).

b. Apply minimum turn-on voltage to the input and measure the dc component of the output circuit voltage drop which shall not exceed the value specified (see 3.1).

4.9.12 Current range (class C, ac relays) (see 3.15). Tests shall be conducted as described in 4.9.10 and 4.9.11 except the output current shall be 1 percent of highest load current rating. The dc offset voltage and waveform distorton shall not exceed specified values (see 3.1).

4.9.13 Zero crossover (class C, ac relays) (when specified, see 3.1) (see 3.16 and 6.8). Apply rated output voltage and resistive load to the relay. Apply rated input voltage. Apply and remove the input control voltage at random and monitor the output for zero voltage turn on and zero current turn off.

4.9.14 Bias current (when applicable) (class C relays) (see 3.17).

a. Apply rated voltages and currents to the output and bias (see 3.1).

b. Read the bias current when the relay is energized with the rated input voltage (relay shall be in on-state).

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4.9.15 Maximum deactuation voltage (class C relays) (see 3.18). Apply minimum specified input voltage to turn the relay on, then reduce this input to the maximum deactuation voltage specified until the relay turns off and observe that the relay has deactuated.

4.9.16 Transients (see 3.19).

4.9.16.1 Voltage (surge) (see 3.19.1). Unless otherwise specified (see 3.1), relays shall be subjected at the input or control terminals to the maximum transient voltage limits for the time specified by MIL-STD-704, limit 1, for category B equipment. No impedance other than that of the wire or cable shall be used between the power source and the input or control terminals for this test.

4.9.16.2 Susceptibility (spike) (see 3.19.2). Relays shall be subjected, at the input or control terminals, to a transient spike voltage of ±600 V, 100 kHz, duration per MIL-STD-704 for 10 applications spaced 10 seconds apart. Relays shall be tested before, during, and after exposure for timing cycle and after exposure for recycle tests (when specified) as specified in 4.9.17 and 4.9.18, respectively.

4.9.16.2.1 Alternate transient susceptibility test. Charge a one microfarad capacitor to 600 V. With a 60 ohm resistor in series discharge capacitor across the input and output control terminals. Five applications spaced 10 seconds between applications. Reverse polarity and repeat five additional applications. Relays shall be tested before, during and after exposure for timing cycle (when specified) and after exposure for recycle tests as specified in 4.9.17 and 4.9.18, respectively.

4.9.16.3 Susceptibility, extraneous voltage (see 3.19.3). With the relay operating voltage applied, apply 25 percent of the nominal control voltage and monitor the output to see that there is no change in condition. Conduct the test at minimum, normal room ambient, and maximum temperature specified.

4.9.16.4 Self-generated (spike) (see 3.19.4) The relay shall be operated normally at nominal voltage. At the input, or control terminals (or both), check for transients when relay is switched on or off and during time interval. The self-generated transient voltage shall not exceed the value specified (see 3.1). (This test to be made without loads on the output controls.)

4.9.17 Timing cycle (when specified) (see 3.20).

CAUTION: If a rise or fall time of the power source is greater than 10 microseconds, the timing cycle may be adversely affected.

4.9.17.1 At 25°C. Relays shall be tested in accordance with the following, as applicable:

a. Time delay on operate: When minimum input voltage is applied to the input terminals, relay must operate within the time interval specified (see 3.1). When voltage is removed, the relay shall return to the deenergized condition within the specified time interval. Output circuits shall be monitored to insure that the switching action is as specified. Repeat test at maximum rated operating voltage.

b. Time delay on release (true): When minimum input voltage is applied to the input terminals, the relay must operate within the specified time interval (see 3.1). When voltage is removed, the relay contacts in classes A, B, D, and E relays, or the solid state output in class C relays, must return to the deenergized condition within the specified time interval. Output circuits shall be monitored to insure that the switching action is as specified. Repeat test at maximum rated operating voltage.

c. Time delay on release (separate control and power terminals): Test as above with minimum and maximum input voltage at both power and control terminals. Repeat test with minimum voltage on power terminals and maximum voltage on control terminals. Repeat again with maximum voltage on power terminals and minimum voltage on control terminals. Time intervals and switching action to be monitored. They must be within values specified (see 3.1).

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d. Interval timer: When minimum or maximum voltages (or both) are applied to the input terminals, switching circuits must operate and return to normal within the time intervals specified (see 3.1). Removal of input voltage shall reset the time delay relay for the next operation, within the reset time specified.

e. Repeat cycle timer or time sequence as specified (see 3.1): When minimum or maximum voltages (or both) are applied to the input terminals, the switching circuits are to operate in accordance with timing cycles and tolerances specified (see 3.1).

4.9.17.1.1 At maximum input voltage. Relays shall be tested in accordance with 4.9.17.1 except at maximum input voltage.

4.9.17.2 High temperature. The relay shall be operated at maximum input voltage at maximum rated ambient temperature for a period of 4 hours for qualification inspection and 45 minutes for quality conformance inspection. Except for indicating means, the output circuit or contacts need not be loaded during this test. Following this period within 1 minute after removal of power, the relay while still maintained at the test temperature shall be tested a specified in 4.9.17.1. The minimum voltage shall be that specified for the high temperature test. The timing period or sequence shall be within the tolerance specified (see 3.1).

4.9.17.3 Low temperature. The relay shall be subjected to the minimum rated ambient temperature for a period of 4 hours for qualification inspection and 45 minutes for quality conformance inspection. At the end of this period and with the relay at the specified temperature, the relay shall meet the test requirements of 4.9.17.1. The timing period or sequence shall be within the tolerance specified (see 3.1). Except for indicating means, the output circuit or contacts shall not be loaded during this test. The indicating means shall not be more than 100 mA or 10 percent of tne rated load, whichever is less.

4.9.18 Recycle time (see 3.21). When tested for timing cycle per 4.9.17.1, or 4.9.17.1.1 as applicable, the power to the input or control terminals as applicable shall be removed and reapplied in the recycle time specified (see 3.1). The relay shall repeat normal operation and the timing cycle shall be within tolerance. The test shall be performed during a timing interval and following a timing interval in accordance with the definitions of recycle time in 6.8.

4.9.19 Reverse polarity (dc operated relays) (see 3.22). Relays shall not be damaged, operate, or start timing when maximum input voltage of reverse polarity is applied to the input or control terminals (or both). Reverse polarity of maximum input voltage shall be applied for a period of not less than 1 minute. Following the test, the delay shall be tested for the timing cycle and recycle time (when specified) as specified in 4.9.17 and 4.9.18 respectively.

4.9.20 Electromagnetic interference (see 3.23). Relays shall be subjected to the electromagnetic interference tests specified in MIL-STD-461, class 1D and MIL-STD-462 with contacts unloaded. Relays shall not be susceptible to electromagnetic interference within the limits specified and shall not create electrical interference in excess of limits specified during timing interval or steady-state operation at nominal rated input voltage (or current).

4.9.21 Crosstalk (when specified, see 3.1) (see 3.24). Crosstalk shall be measured using equipment which shall have an input impedance of 1 megohm, minimum, and shall be paralleled with a capacitance of 20 picofarads maximum. A 1.0 to 10.0 volts peak-to-peak input signal at frequencies up to 10 megahertz shall be applied to the switching circuit through coaxial cable, terminated in 50 ohms ±5 percent at the device terminal. The coaxial cable shall conform to MIL-C-17. The input signal amplitude shall be measured at the terminals of the relay. The terminals where crosstalk is to be detected, shall be connected through a similar type coaxial cable, terminated in 50 ohms ±5 percent at the measuring device. The resultant attenuation, in decibels (dB) equals:

$$20 \left(\text{Log}_{10} \frac{V_{IN}}{V_{OUT}}\right)$$

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4.9.22 Capacitance (when specified, see 3.1) (see 3.25). Relays shall be tested as specified in method 305 of MIL-STD-202. The following detail and exception shall apply:

a. Test frequency: 1 kHz, unless otherwise specified (see 3.1).

b. Points of measurement as specified, see 3.1.

4.9.23 Continuous current (see 3.26). Relays shall be energized continuously for 100 hours with maximum input voltage applied to the input or control terminals (or both), and with maximum rated load connected to each output circuit. The test shall be performed at the maximum temperature and altitude specified (see 3.1). Following the test and with the relay at the maximum temperature and altitude, and contacts still loaded, the relay shall be deenergized and shall immediately be tested for timing cyle and recycle time as specified in 4.9.17 and 4.7.18, respectively. The minimum voltage shall be as specified for the continuous current test (see 3.1). After the test the relay shall be tested for insulation resistance, dielectric withstanding voltage, static contact resistance (or voltage drop) as specified in 4.9.5., 4.9.6, and 4.9.7, respectively.

4.9.24 Resistance to soldering heat (not applicable to plug-in relays) (see 3.27). Relays shall be tested in accordance with method 210 of MIL-STD-202. The following details shall apply:

a. Depth of immersion in molten solder: Within .060 ±.020 inch (1.52 ±0.51 mm) of the relay base.

b. Test condition letter B.

c. Measurements after test: Insulation resistance, dielectric withstanding voltage, timing cyle, and recycle time (when specified) shall be measured as specified in 4.9.5, 4.9.6.1, 4.9.17, and 4.9.18, respectively.

4.9.25 Salt spray (corrosion) (see 3.28). Relays shall be tested in accordance with method 101 of MIL-STD-202. The following detail and exceptions shall apply:

a. Applicable salt solution: 5 percent.

b. Test condition letter B.

c. Inspection after test: Relays shall be inspected for evidence of peeling, chipping, blistering of the finish, or exposure of base metal due to corrosion.

d. Measurement after test: Timing cycle and recycle time (when specified) shall be measured as specified in 4.9.17 and 4.9.18.

4.9.26 Thermal shock (see 3.29). Relays shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

a. Test condition letter: As specified (see 3.1).

b. Measurements after cycling: Relays shall be visually inspected for cracking, peeling and flaking of the finish, and the insulation resistance, dielectric withstanding voltage, static contact resistance (or voltage drop), timing cycle and recycle time (when specified) shall be measured as specified in 4.9.5, 4.9.6.1, 4.9.7.1, 4.9.17 and 4.9.18, respectively.

4.9.27 Low temperature operation (see 3.30). The relay shall be subjected to the low temperature specified (see 3.1), for a period of 48 hours. At the end of this period, and with the relay at the low temperature, the timing cycle and recycle time

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(when specified), and static contact resistance (or voltage drop) shall be measured as specified in 4.9.17, 4.9.18, and 4.9.7, respectively. (These tests shall be accomplished in the sequence listed and in a minimum amount of time to prevent significant heating of the coil.) The relay shall then be tested intermittently for timing cycle and recycle time until it attains room temperature. Relays which contain permanent magnets in the magnetic circuit, shall, in addition to the above tests, be subjected to the demagnetizing effect of the cold coil energized with maximum voltage specified. During the low temperature test, after approximately 24 hours, these relays shall be operated by the sudden application of maximum input voltage for one operation. All units subjected to this demagnetizing effect shall be tested in accordance with high temperature timing cycle (see 4.9.17.2), at the conclusion of this test.

4.9.28 Shock (specified pulse) (see 3.31). Unless otherwise specified (see 3.1), relays shall be tested in accordance with method 213 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting method: Rigidly mounted by normal mounting means.

b. Test condition letter: As specified (see 3.1).

c. Electrical-load conditions: In each direction of shock, the relay shall be deenergized during two shocks and energized with maximum input voltage or current during one shock.

d. Tests during shock: Contacts shall be monitored with an adequate test circuit as approved by the qualifying activity. The test circuit shall verify that no false closure or abnormal bridging of open contacts occurs in excess of one microsecond.

e. Inspection after test: Insulation resistance, dielectric withstanding voltage, static contact resistance (or voltage drop), timing cycle and recycle time (when specified), in 4.9.5, 4.9.6.1, 4.9.7, 4.9.17 and 4.9.18, respectively. Relays shall then be inspected for loosening of parts.

4.9.29 Vibration, high frequency (see 3.32).

4.9.29.1 Vibration, high frequency (classes A and B). Unless otherwise specified (see 3.1), relays shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting method: Rigidly mounted by normal mounting means. Connections to the relay shall be made by soldering flexible standard wires to the relay terminals.

b. Electrical-load conditions: Relays shall be tested with the coil energized for 2 hours at maximum input and with the coil deenergized for 2 hours, in each of the three mutually perpendicular directions. Except as specified in d., contacts shall not be loaded.

c. Test condition letter D, except frequency shall be 5 Hz to 2,000 Hz.

d. Tests during vibration: As specified in 4.9.28.d.

e. Inspections after test: As specified in 4.9.28.e.

f. For group A testing (applicable only to failure rate levels "M", "P" and "R" only), only one cycle shall be performed. The relay shall be vibrated in each of three mutually perpendicular directions. Relays having both normally open and normally closed contacts shall be subjected to one vibration scan in the operating mode and one scan in the non-operating mode. Contact monitoring shall be in accordance with 4.9.28.d.

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4.9.29.2 Vibration, high frequency (class C). Unless otherwise specified (see 3.1), relays shall be tested in accordance with method 204 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting method: Rigidly mounted by normal mounting means. Connections to the relay shall be made by soldering flexible standard wires to the relay terminals.

b. Electrical-load conditions: Relays shall be tested, energized for 2 hours at rated voltage and deenergized for 2 hours, in each of the three mutually perpendicular directions.

c. Test condition letter F.

d. Inspections after test: As specified in 4.9.28.e.

4.9.29.3 Vibration random. When specified (see 3.1), relays shall be tested in accordance with method 214 of MIL-STD-202. The following details shall apply:

a. Mounting method: Rigidly mounted by normal mounting means. Connections to the relay shall be made by soldering flexible standard wires to the relay terminals.

b. Electrical-load conditions: Relays shall be tested with the coil energized for 7.5 minutes at maximum input voltage and with the coil deenergized for 7.5 minutes, in each of the three mutually perpendicular directions. Except as specified in d., contacts shall not be loaded.

c. Test condition I, letter G, duration 15 minutes.

d. Measurements during vibration: As specified in 4.9.28.d.

e. Inspections after test: As specified in 4.9.28.e.

4.9.30 Acceleration (see 3.33). Relays shall be tested in accordance with method 212 of MIL-STD-202. The following details and exceptions shall apply:

a. Mounting: The relays shall be rigidly fastened, using normal mounting means, in each of three mutually perpendicular positions.

b. Test condition letter A, except that acceleration shall be applied in each direction along three mutually perpendicular axes of the specimen and one of the axes shall be in the direction most likely to cause malfunction. Unless otherwise specified (see 3.1), the acceleration force shall be 20 g's.

c. Electrical operating conditions during acceleration: Relays shall be tested with the relay in deenergized condition and repeated with the relay coil energized at specified 25°C pickup voltage. An indicating instrument shall be connected across the contacts while the relay is undergoing this test to determine the ability of the contacts to remain in the proper position. Open contacts shall be wired in parallel and closed contacts shall be wired in series during this test.

d. Tests during acceleration: Timing cycle and recycle time (when specified) as specified in 4.9.17 and 4.9.18, respectively.

4.9.31 Resistance to solvents (see 3.34). Relays shall be tested in accordance with method 215 of MIL-STD-202. The following details and exception shall apply:

a. Portion of the specimen to be brushed: All marking.

b. Number of specimens to be tested: Two using first solvent solution; and one specimen each using second and third solvent solutions. A total of four specimens shall be tested.

c. Specimen shall be inspected for legibility of marking.

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oscilloscope with a rise time of 1020 microseconds

4.9.32 Strength of terminals and mounting studs (see 3.35).

4.9.32.1 Strength of threaded terminals and mounting studs. Relays having mounting studs or threaded terminals, shall be tested to determine compliance with the strength requirements specified in 3.35.1, and tables II and III. The specified pull force shall be applied both coaxially with the threaded terminal in a direction away from the main body of the relay, and again normal to the threaded axis of the terminal in approximately the same plane as the seat for the terminal lug. The specified torque shall be applied to the terminal mounting nut, or screw with all terminal mounting hardware, including one terminal lug of proper size, assembled in proper order. The relays shall withstand the specified force for 1 minute without damage. Relays, with threaded terminal assemblies soldered in place, shall be tested as follows (not applicable to relays with maximum temperature rating above 125°C). The relays shall be stabilized in a 180°C ambient temperature. After stabilizing at this temperature for a minimum of 2 hours, the terminals shall be subjected to the pull and torque test as specified above.

4.9.32.2 Strength of solder-lug, plug-in, and wire-lead terminals (see 3.35.2). Relays shall be tested in accordance with method 211 of MIL-STD-202 in accordance with the following, as applicable. Unless otherwise specified herein, two terminals of each discrete design, size, and configuration shall be tested; however, if there is only one of such design, size and configuration, it shall be tested.

4.9.33 Overload.

4.9.33.1 Overload (applicable to high level relays only (see 3.36.1). Relay contacts (both normally open and normally closed) shall be subjected to the tests specified in 4.9.33.1.1 through 4.9.33.1.3, as applicable. For double-throw relays, separate tests shall be performed for the normally open and the normally closed contacts. Relay coil energization shall be nonsynchronous with the power supply for ac loads. The number of operations shall be 100 for dc contact loads, and 200 for ac contact loads ("on" and "off" times shall be approximately equal). The coil shall be energized at rated voltage. Overload current shall be twice rated load current. The opening and closing of all contacts shall be continuously monitored to detect switching of the contacts. The equipment shall be capable of monitoring 50 percent of the actual opened and closed time of each contact. The equipment shall automatically cut off when a failure occurs, or shall record any failures. The relay case shall be connected to system-ground through a normal-blow fuse rated at 5 percent of the test current or 100 mA, whichever is greater. After the test, the fuse shall be tested for electrical continuity.

4.9.33.1.1 Resistive load. Suitable noninductive resistors shall be used. Cycling rate shall be 20 ±2 cycles per minute (cpm).

4.9.33.1.2 Inductive. Cycling rate shall be 10 ±1 cpm.

4.9.33.1.2.1 Inductive, dc. The L/R ratio shall be the same as for rated load (see 3.1).

4.9.33.1.2.2 Inductive, ac. The load shall consist of inductive and resistive elements with 0.7 ±0.05 lagging power factor at the voltage and frequency specified (see 3.1).

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4.9.34 Life (see 3.37) (class A and class B relays). Unless otherwise specified (see 3.1), relays shall be cycled for 100,000 operations with contacts loaded in accordance with 4.8.34.1 through 4.8.34.5, as applicable. Relay coil energization shall be nonsynchronous with the power supply for ac loads. Each relay case shall be connected to system ground through an individual normal-blow fuse rated at the greater of 100 mA or 5 percent of load current. Ambient temperature of relays shall be 125°C minimum. Each contact shall be monitored during each operation in accordance with 4.9.2.4.1.1. All contacts in each sample unit shall switch identical loads. Following cycling, the electrical continuity of each case-grounding fuse shall be checked.

4.9.34.1 Resistive load. Cycling rate shall be 20 ±2 operations per minute. Suitable resistors shall be used. Current shall be rated resistive current as specified (see 3.1).

4.9.34.2 Inductive load. Current shall be rated current. Appropriate inductive load components (see 4.9.31.2.1 and 4.9.31.2.2, as applicable) shall be used. A suitable resistor may be placed in the circuit to obtain rated steady-state current flow. Cycling rate shall be 10 ±1 cpm with equal "on" and "off" periods.

4.9.34.3 Lamp load. Unless otherwise specified, the lamp load shall be performed with the 28 V dc power supply voltage. Relays shall be subjected to making 12 times the rated lamp load and breaking the rated lamp load. The duration of the 12 times inrush shall be 0.015-0.020 second, the total "on" time shall be 5 ±0.05 seconds and the "off" time shall be 5 ±2 seconds.

4.9.34.4 Low level and mechanical life. Relays shall be cycled 1,000,000 (class A) and 400,000 cycles (class B) operations. The first 100,000 operations only are to be monitored and used for failure rate level determination. The minimum cycle time shall be 10X the sum of the maximum operate and release times for the relay under test. Each contact load shall be 10 to 50 microamperes at 10 to 50 mV (dc or peak ac). The coil shall be energized at rated voltage during 50 ±10 percent of each operation. The remaining 900,000 operations need not be monitored.

4.9.34.5 Life (class C relays only). Relays shall be subjected to the following procedure:

a. With the relays mounted in accordance with 4.9.2.3a, they shall be operated for 48 hours in accordance with 4.9.2.5 except the duration of the turn on shall be 50 ±5 percent of each operation of the relay. The ambient temperature of the relay environment shall be maintianed at the highest specified operating temperature (see 3.1). Repeat as above for 48 hours, except with the ambient temperature at the lowest specified operating temperature. Repeat as above for 48 hours, except with the temperature at room ambient.

b. Same as subparagraph a., except with the maximum turn-off voltage continuously applied to the input. Relays shall be monitored to verify that they remain turned off.

c. Same as b. above, except with the minimum turn-on voltage continuously applied to the input. Relays shall be monitored to verify that they remain turned on.

d. The transient voltage test shall then be performed as specified in 4.9.16.1.

e. Repeat steps a. through d. above for each additional load specified.

f. After the above tests, relays shall be subjected to the insulation resistance, dielectric withstanding voltage, and all the electrical tests in accordance with 4.9.5, 4.9.6, 4.9.8 through 4.9.15, respectively.

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4.9.35 Intermediate current (see 3.38). Relays shall be tested as specified in method 312 of MIL-STD-202. The following details and exceptions shall apply:

a. Maximum contact resistance: As specified in 3.10.1.

b. Coil energizing voltage: Maxinum input voltage (see 3.1).

c. Cycling rate: 10 ±2 cycles per minute with minimum of 75 percent coil "on" time.

d. Contact current/voltage: 100 mA at 28 V dc.

e. Monitoring: At least 40 percent of the closed time of each contact during each cycle shall be monitored for contact resistance.

f. Number of cycles: 50,000 cycles.

g. Ambient temperature: +125°C (unless otherwise specified).

5. PACKAGING

5.1 Preservation. preservation shall be level A, B or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Relays shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. Relays shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packs. Each relay shall be unit packed one each in accordance with method III of MIL-P-116 insuring compliance with the applicable requirements of that specification. Terminals shall be protected by container design, die-cast inserts or other suitable noncorrosive supporting materials or devices. The container shall conform to variety 2 of PPP-B-566 of PPP-B-676.

5.1.1.5 Intermediate packs. Intermediate packs are not required.

5.1.2 Level B. The requirements for level B shall be as specified for level A except that any variety of the unit containers specified may be used.

5.1.3 Level C. The level C preservation of relays shall conform to the MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Relays, preserved as specified in 5.1, shall be packed in wood boxes conforming to PPP-B-601, overseas type or PPP-B-621, class 2. Closure and strapping shall be in accordance with the applicable container specification except that metal strapping shall conform to QQ-S-781, type I, finish A. The requirements for level B packing shall be used when the total quantity of a stock numbered product for a single destination does not exceed a packed volume of one cubic foot.

5.2.2 Level B. Relays, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PP-B-636, class weather resistant, style optional, special requirements. The requirements for box closure, waterproofing and reinforcing shall be in accordance with method V of the PPP-B-636 appendix.

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rated operating voltage.

5.2.3 Level C. Relays, preserved as specified in 5.1, shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. Relays, packed as specified in 5.2.1, shall be unitized on pallets in conformance with the MIL-STD-147, load type I, with a wood cap (storage aid 5) positioned over each load.

5.2.4.2 Level B. Relays, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that weather resistance fiberboard caps (storage aid 4) shall be used in lieu of wood caps.

5.2.4.3 Level C. Relays, packed as specified in 5.2.3, shall be unitized as specified in 5.2.4.2 except that the fiberboard caps shall be class domestic.

5.3 Marking. In addition to any special or other identification marking required by the contract (see 6.2), each unit and exterior container and unitized load shall be marked in accordance with MIL-STD-129. The complete military or contractor's type or part number, as applicable (including the FSCM), shall be marked on all unit and supplementary packs in accordance with the identification marking provisions of MIL-STD-129.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2 and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.8.3.

5.4.3 Army acquisitions.

5.4.3.1 Levels A and B packing. When the gross weight exceeds 200 pounds or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification. Unitization shall be required when the containers specified in 5.2.1 and 5.2.2 do not require skids, quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet, and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified, except that container strapping may be omitted, shall be placed on a pallet, load type I conforming to MIL-STD-147. For level B, unit containers which meet these requirements may be palletized without further packing. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV. Stretch wrap in accordance with MIL-STD-147 is authorized for shipments within the continental United States and for containerized shipments.

6. NOTES

6.1 Intended use. Relays conforming to this specification are intended for use in dc or ac (single or polyphase) electrical systems as a means of sensing frequencies, voltages, phases and currents to protect the operated equipment devices. Their principal areas of application (i.e., aircraft, missiles, spacecraft, and ground support equipment) does not preclude the use of these relays in other military applications.

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6.1.1 Contact rating. The contact ratings of relays covered by this specification are based on load endurance tests which establish the relay capability to switch rated loads.

6.1.2 Failure rate level. Failure rate level (percent per 10,000 operations) as specified in table I, is based on an average of 10 operations per hour, per application, for 1,000 hours in accordance with MIL-STD-690.

6.1.3 Packaging requirements. The preservation, packing and marking specified herein are intended for direct shipments to the Government. However, at the option of the contractor or when so specified, the packaging provisions herein are also applicable for the preparation of relays for shipment from the parts contractor to the original equipment manufacturer.

6.2 Ordering data.

a. Title, number, and date of this specification.

b. Title, number, and date of the applicable specification sheet.

c. Levels of preservation and packing required (see 5.1 and 5.2).

d. If special or additional identification marking is required see 5.3.

e. Marking requirements for failure rate level (see MIL-STD-690).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 39016 whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the QPL is the Naval Electronics Systems Command, ELEX 8111, Department of the Navy, Washington, DC 20363, however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center (DESC), Dayton, OH 45444-5000. Application for qualification tests shall be made in accordance with provisions governing qualification SD-6 (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Conditions for use of level B preservation. When level B preservation is specified (see 5.1.2), this degree of protection should be used for the acquisition of relays for resupply worldwide under known favorable handling, transportation and storage conditions.

6.5 Sealing. The following procedure is suggested as a guide; however, every effort should be made to utilize the most effective procedure consistent with the state of the art.

a. Evacuate to less than 200 microns.

b. Heat to maximum rated ambient temperature with continued evacuation (see 3.1).

c. Maintan heat and vacuum for 12 hours or longer, continuing the treatment until a maximum pressure of 80 microns is reached.

d. Turn off heaters and maintain pressure for 4 hours.

e. Close evacuation valve and fill chamber with the desired pressurizing gas.

f. Seal relay before removing from chamber.

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6.6 Intermetallic contact. The finishing of metallic areas to be placed in intimate contact by assembly presents a special problem, since intermetallic contact of dissimilar metals results in electrolytic couples which promote corrosion through galvanic action. To provide the required corrosion protection, intermetallic couples are restricted to those permitted by table XIV. Table XIV shows metals and alloys (or plates) by groups which have common electromotive forces (EMF) within 0.05 volt when coupled with a saturated calomel electrode in sea-water at room ambient temperatures. All members of a group are considered as completely compatible, one with the other. Compatible couples between groups have been specified in table XIV based on a potential difference of 0.25 volt maximum. To simplify any arithmetic involved, table XIV shows, in addition to EMF against a calomel electrode, a derived "anodic index" with group 1 (gold etc) as 0 and group 18 (magnesium, etc.) as 175. Subtraction of a lower group anodic index gives the EMF difference in hundredths of a volt.

6.6.1 Groups. Table XIV sets up 18 primary groups. It may be noted that neither the metallurgical similarity or dissimilarity of metals is the parameter for selection of compatible couples. All members within a group, regardless of metallurgical similarity, are considered inherently nonsusceptible to galvanic action, when coupled with any member within the group; for example, such dissimilar metals as platinum and gold. Similarly, such basically dissimilar alloys as austenitic stainless steel, silver-solder, and low brass (all members of group 5) are inherently nonsusceptible when coupled together.

6.6.2 Compatibility graphs. Permissible couple series are shown in table XIV by the graphs at the right. Members of groups connected by lines will form permissible couples. A O indicate the most cathodic member of each series, a o an anodic member, and the arrow indicates the anodic direction.

6.6.3 Selection of compatible couples. Proper selection of metals in the design of equipment will result in fewer intermetallic contact problems. For example, for sheltered exposure, neither silver nor tin require protective finishes. However, since silver has an anodic index of 15 and tin 65, the EMF generated as a couple is 0.50 volt, which is not allowable by table XIV. In this case, other metals or plates will be required. It should be noted that in intermetallic couples, the member with the higher anodic index is anodic to the member with the lower anodic index and will be susceptible to corrosion in the presence of an electrolytic medium. If the surface area of the cathodic part is significantly greater than that of the anodic part, the corrosive attack on the contact area of the anodic part may be greatly intensified. Material selection for intermetallic contact parts, therefore, should establish the smaller part as the cathodic member of the couple, whenever practicable.

6.6.4 Plating. When base metals intended for intermetallic contact form couples not allowed by table XIV, they are to be plated with those metals which will reduce the potential difference to that allowed by table XIV.

6.7 Magnetic orientation. Magnetic orientation of a relay may be determined by energizing the coil with a know polarity and checking for attraction or repulsion of a permanent magnet to an external surface of the relay.

6.8 Glossary. The definitions listed below are not a complete glossary of relay terminology, but rather are intended as definitions of the technical terms as applied within the specification.

Adjunct sealant: Any hydrocarbon material used on the exterior of the unit to improve the hermeticity of a hermetically sealed relay.

Bifilar winding: Two or more windings with the wire of each winding alongside the other, matching turn for turn; may be either inductive or noninductive.

Bounce time: The interval between first make of the contact until the uncontrolled making and breaking of the contact ceases.

Chatter, contact: The undesired opening of mating contacts resulting from uncompensated ac operation, or from external shock and vibration.

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Classes. Relays classes are defined as follows:

a. Classes A and B (hybrid): A combination of solid state circuit input elements function with an electromagnetic relay that performs the switching function or time delay or both.

b. Class C (solid state): A combination of electronic devices such as semiconductors, resistors, capacitors, etc., assembled to perform as a sensing device.

Coil: One or more windings on a common form.

Contacts: The current-carrying parts of a relay that open or close electrical circuits.

Contact, bifurcated: A forked or branched, contacting member so formed or arranged, as to provide dual contacting.

Contact, closed: A normally-closed contact with the relay released is a closed contact. A normally-open contact with the relay operated is a closed contact.

Contact, open: A normally-closed contact with the relay operated is an open contact. A normally-open contact with the relay released is an open contact.

Contact arrangement: The combination of contact forms that make up the entire relay switching structure.

Contact bounce: Intermittent opening of contacts after initial closure due to contact impact.

Contact force: The force exerted by a movable contact against a mating contact when the contacts are closed.

Contact gap: The distance between a pair of mating relay contacts when the contacts are open.

Contact weld: A contact failure due to fusing of closed contacts, resulting in their failure to open.

Degassing: A process of thermally cleaning a relay (at elevated temperatures) to drive off harmful gasses. Degassing is usually performed using vacuum pumps.

Dropout current or voltage: The current or voltage at which the last normally closed contact closes.

Hermetically sealed relay: A relay contained within an enclosure that is sealed by fusion to insure a low rate of gas leakage. This usually refers to metal-to-metal, or metal-to-glass sealing.

Nonoperating current or voltage: The maximum coil current or voltage at which a relay remains in the deenergized condition.

Recycle time:

a. Time delay on operate and interval time relays: The minimum time that power must be removed from the input terminals in order to insure that a new timing cycle can be initiated within the specified timing tolerance.

b. Time delay on release relays: The minimum time that power must be applied to the input or control terminals in order to insure that the timing cycle can be completed as specified.

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FIGURE XIV. Compatible couples (see 6.5).1/

Group No.	Metallurgical category	EMF (volt)	Anodic index (0. 01 v)	Compatible couples
1	Gold, solid and plated; gold-platinum alloys; wrought platinum (most cathodic)	+0.15	0	
2	Rhodium plated on silver-plated copper	+0.05	10	
3	Silver, solid or plated; high silver alloys	0	15	
4	Nickel, solid or plated; monel metal, high nickel-copper alloys	-0.15	30	
5	Copper, solid or plated; low brasses or bronzes; silver solder; German silver; high copper-nickel alloys; nickel-chromium alloys; austenitic corrosion-resistant steels	-0.20	35	
6	Commercial yellow brasses and bronzes	-0.25	40	
7	High brasses and bronzes; naval brass; Muntz metal	-0.30	45	
8	18 percent chromium type corrosion-resistant steels	-0.35	50	
9	Chromium, plated; tin, plated; 12 percent chromium type corrosion-resistant steels	-0.45	60	
10	Tin-plate; terneplate; tin-lead solder	-0.50	65	
11	Lead, solid or plated; high lead alloys	-0.55	70	
12	Aluminum, wrought alloys of the duralumin type	-0.60	75	
13	Iron, wrought, gray, or malleable; plain carbon and low alloy steels, armco iron	-0.70	85	
14	Aluminum, wrought alloys other than duralumin type; aluminum, case alloys of the silicon type	-0.75	90	
15	Aluminum, cast alloys other than silicon type; cadmium, plated and chromated	-0.80	95	
16	Hot-dip-zinc plate; galvanized steel	-1.05	120	
17	Zinc, wrought; zinc-base die-casting alloys; zinc, plated	-1.10	125	
18	Magnesium and magnesium-base alloys, cast or wrought (most anodic)	-1.60	175	

1/ Compatible couples - potential difference of 0.25 volt maximum between groups.

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Types. Sensor types are defined as follows:

a. Type I: AC power monitoring sensors. These types of sensors are a combination of voltage sensor, frequency sensor, and phase sequence sensor in the same package.

b. Type II: DC voltage sensors. Relay types which sense dc voltage level.

c. Type III: AC voltage sensors. Relay types which sense ac voltage level.

d. Type IV: Frequency sensors. Relay types which sense the frequency of the source.

e. Type V: Phase sequence sensors. Relay types which sense the phase sequence of a three phase line.

Zero voltage turn on and zero current turn off. A characteristic that requires the relay to turn "on" and turn "off" only at the half-cycle, zero crossing point when connected to a load, regardless of when the control voltage is applied or removed.

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Air Force - 85

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